



传统 中医 药物 毒性 预测 通过 异质 网络

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文章 信息

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摘要

clinical usage of Traditional Chinese Medicines (TCMs) gains increasing international attention for their distinct therapeutic effects. Precisely understanding the herb (TCM) toxicity and identifying toxic and effective herb ingredients are crucial for their safe use. Traditional wet-lab based pipelines for assessing herb toxicity are complex and time-consuming. Given the large volume toxicity data of ingredients, building computational models for efficient ingredient-based toxicity prediction and evaluation is promising, but often fails to effectively predict the toxicity of herbs due to the complexity herbs and toxic mechanisms. To address these challenges, we propose a heterogeneous network based approach (HerbToxNet) for predicting herb toxicity. HerbToxNet first constructs a heterogeneous network composed with herbs, ingredients and molecular targets, and leverages a heterogeneous graph attention network to learn the representation of herbs. Meanwhile, it performs contrastive learning with dynamic coefficient to refine the representation by pulling close the herbs with shared toxic labels, while pushing away the others. Additionally, we introduce a weighted label fusion strategy that uses toxic labels of similar herbs to augment the predicted labels of this herb. HerbToxNet outperforms competitive methods and finds out novel potential toxicities. It uses Multilayer Perceptron (MLP) on the herb representation to predict the toxic labels of this herb. It can mine related toxic ingredients and targets in an interpretable way, and dissect the molecular mechanism of herb toxicity with authenticity.

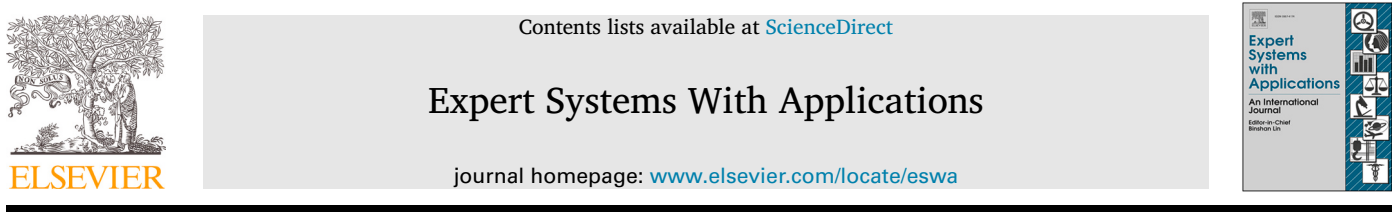
1. 引言

草药医学在全球传统医疗实践中有着悠久历史，尤其在亚洲地区，数个世纪以来一直是医疗保健的基石（Hu et al., 2025）。草药含有多种生物活性成分，如生物碱、黄酮类化合物和萜烯类，这些成分共同发挥治疗作用。尽管被认为安全天然，但许多草药已被证实存在不良反应，包括肝毒性、肾毒性、心脏毒性等（Jiang et al., 2010）。这些毒性效应可能由不当剂量、长期使用、草药相互作用或重金属与农药污染引起。此外，因栽培、采收和加工方法差异导致的草药质量波动，更增加了安全性评估的复杂性（Liu et al., 2020）。面对这些多维度挑战，亟需建立高效

可靠 的 系统性 方法 来 识别 与 评估 草药 毒性。

与化学药物的单一成分不同，草药含有多种生物活性成分，具有多靶点、多通路和多效应相互作用的特点，其毒性机制更为复杂多样（Wu等人，2021）。因此，草药毒性研究面临若干独特挑战。首先，草药毒性研究需要复杂且成本高昂的实验。不同文献数据常存在显著异质性和不一致性，而临床数据分散在各机构，缺乏标准化格式和共享平台，这使得获取高质量毒性数据变得困难（Chan，2003）。其次，草药的多成分特性和复杂毒性机制对模型提出更高要求。然而，传统机器学习和深度学习方法难以处理草药的多模态特征（Wang等人，2004）。第三，草药

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Traditional Chinese medicine toxicity prediction by heterogeneous network

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A R T I C L E I N F O

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Contrastive learning

A B S T R A C T

The clinical usage of Traditional Chinese Medicines (TCMs) gains increasing international attention for their distinct therapeutic effects. Precisely understanding the herb (TCM) toxicity, and identifying toxic and effective herb ingredients are crucial for their safe use. Traditional wet-lab based pipelines for assessing herb toxicity are complex and time-consuming. Given the large volume toxicity data of ingredients, building computational models for efficient ingredient-based toxicity prediction and evaluation is promising, but often fails to effectively predict the toxicity of herbs, due to the complexity herbs and toxic mechanisms. To address these challenges, we propose a heterogeneous network based approach (HerbToxNet) for predicting herb toxicity. HerbToxNet first constructs a heterogeneous network composed with herbs, ingredients and molecular targets, and leverages a heterogeneous graph attention network to learn the representation of herbs. Meanwhile, it performs contrastive learning with dynamic coefficient to refine the representation by pulling close the herbs with shared toxic labels, while pushing away the others. Next, it uses Multilayer Perceptron (MLP) on the herb representation to predict the toxic labels of this herb, and further introduces a weighted label fusion strategy that uses toxic labels of similar herbs to augment the predicted labels of this herb. HerbToxNet outperforms competitive methods and finds out novel potential toxicities, with 96% toxicity labels confirmed for canonical herbs. It can mine related toxic ingredients and targets in an interpretable way, and dissect the molecular mechanism of herb toxicity with authenticity.

1. Introduction

Herb medicine (Herb) has a long history in traditional medical practices around the world, especially in Asia, where it has been a cornerstone of healthcare for centuries (Hu et al., 2025). Herbs consist of a variety of bioactive ingredients, such as alkaloids, flavonoids, and terpenes, which contribute to their therapeutic effects. Although they are considered as safe and natural, many herbs have been linked to adverse effects, including hepatotoxicity, nephrotoxicity, cardiotoxicity, etc. (Jiang et al., 2010). These toxic effects can be caused by improper dosing, prolonged use, herb interactions, or contamination with heavy metals and pesticides. In addition, variations in herb quality arising from differences in cultivation, harvesting, and processing methods cause further complexity into their safety evaluation (Liu et al., 2020). Given these multifaceted challenges, there is an urgent need to develop effi-

cient and reliable methodologies to systematically identify and evaluate herb toxicity.

Different from the simple composition of chemical drugs, a herb contains a variety of bioactive ingredients with multi-target, multi-pathway and multi-effect interactions, and its toxicity mechanisms are more complex and diverse (Wu et al., 2021). Therefore, the study of herb toxicity faces some distinctive challenges. First, herb toxicity studies require complex and costly experiments. Data from different literature often exhibit substantial heterogeneity and inconsistency, while clinical data are scattered across institutions, lacking standardized formats and shared platforms. This makes it difficult to obtain high-quality toxicity data (Chan, 2003). Second, the multi-ingredient nature and complex toxicity mechanisms of herbs impose higher demands on models. However, traditional machine learning and deep learning methods struggle to process the multi-modal features of herbs (Wang et al., 2004). Third, herb

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毒性预测常被简化为二元分类任务，但其本质应是多标签分类任务，因为一种草药往往关联多种毒性效应。在我们收集的中药毒性数据集中，超过半数的草药（252种中的141种）关联两个及以上毒性标签，这凸显了基于多标签学习的草药毒性预测的复杂性与必要性。第四，深度学习模型常被视为“黑盒”，难以对其毒性预测提供清晰的生物学解释（Gao 等人，2025）。这种可解释性的缺失阻碍了研究者对毒性机制的理解，从而影响深度模型的真实性和临床适用性。

监管机构与研究人员一直强调加强草药产品安全与质量控制的重要性（Jit̃areanu 等人，2022）。传统的草药毒性评估方法（如体内动物试验和体外细胞试验）不仅耗时昂贵，还引发伦理问题。此外，鉴于成分数量庞大及其作用机制多样，全面毒性评估仍是一项艰巨任务。随着高通量技术的出现和大规模生物与化学数据集的日益可用，计算方法在草药毒性预测领域备受关注。这些方法包括随机森林、支持向量机等机器学习模型以及深度学习框架，能够通过分析复杂的多维数据来预测毒性结果（He 等人，2019a; Sun 等人，2019; Yang 等人，2021）。通过整合化学结构、药理活性和已知毒理学特征的数据，这些模型为识别草药潜在毒性并促进其安全使用提供了经济高效的可扩展解决方案（Jia 等人，2024; Wang 等人，2020; Wu等人，2019a; Yu等人，2025）。然而，现有计算方法仍面临由草药毒性数据异质性与不完整性的重大挑战。当前大多数模型仅局限于单一信息类型（如分子结构或药理活性），且往往缺乏有效整合反映复杂草药生物学机制的多样化数据源的能力。此外，由于缺乏能够捕捉草药、成分、靶点与毒性间多层次相互作用的系统框架，毒性预测的可解释性与真实性均受到限制。鉴于此，亟需开发能够系统融合多源信息并解决草药毒性预测中数据异质性、稀疏性与可解释性挑战的可靠计算方法。

本文通过整理现有草药毒性研究结果，构建了一个多标签毒性数据集以解决草药毒性数据分散稀缺的问题（见第4节）。为解决复杂草药建模困难及缺乏可解释性的问题，我们提出了一种基于异质网络的方法（HerbToxNet）用于草药毒性预测。HerbToxNet首先利用药理学特性、毒理学特征及草药间关联构建草药-成分-靶点异质网络，随后采用异质图注意力网络（Wang等人，2019年）学习草药表示。同时引入带动态系数的对比学习，使具有共同毒性标签的草药对表示更接近，而不同毒性标签的草药对表示更疏远。接着使用多层感知器对草药表示进行初始毒性标签预测，并进一步采用加权标签融合策略，通过检索相似草药并融合其毒性标签来优化预测标签。

Our contributions are summarized as below:

(i) 我们构建并共享一个多标签毒性数据集，通过整理分散在不同草药数据库和已发表文献中现有草药毒性研究的结果。

(ii) 我们提出一个统一框架 HerbToxNet，通过异质网络建模草药、成分与靶点间的多层次关系，实现对毒性相关机制的透明分析并增强可解释性。

(iii) HerbToxNet 融合异质网络建模、对比学习与加权标签融合，以应对毒性数据稀缺问题，实现符合逻辑且真实可信的毒性预测。

(iv) 广泛实验表明，HerbToxNet 优于竞争性方法，能够识别经典草药的潜在毒性（96% 已确认），并且可以直接从可解释视角揭示与毒性相关的成分和靶点。

本文其余部分组织结构如下：第2节讨论相关工作，第3节详细阐述所提出的HerbToxNet框架，第4节介绍实验设置、结果与分析，第5节总结我们的工作并展望未来研究方向。

2. 相关工作

现有的草药毒性计算方法通常预测一些主要毒性类型，如肝毒性、肾毒性、心脏毒性等，并将其建模为二分类任务（关注毒性是否存在）。基于草药毒性的特点，已发展出一系列草药毒性预测理论和方法，可分为三类：基于定量构效关系（QSAR）的草药毒性预测、网络毒理学和药性理论。

在计算毒理学中，常通过计算方法利用化合物的虚拟结构来分析、模拟、可视化或预测化学物质的毒性（Raies & Bajic, 2016）。定量构效关系能够定量捕捉分子结构与生物效应之间的关系，它赋能了化合物毒性的计算预测，并已广泛应用于化学药物的毒性评估（Cherkasov et al., 2014）。因此，基于QSAR的方法可适用于计算预测每种草药内单个成分（类似药物）的毒性，为评估草药整体毒性提供基础（Gao 等人，2019）。He 等人（2019a）收集肝毒性草药及成分，分析这些成分的结构后发现生物碱和萜类化合物是主要肝毒性成分。Yang 等人（2021）运用定量构效关系评估决明子中成分的潜在毒性，通过线性模型阐明结构特征与细胞毒性的关系，发现决明子中三种成分具有肝毒性，八种成分具有肾毒性。Sun 等人（2019）利用草药肾毒性数据构建定量构效关系模型，并采用人工神经网络和支持向量机算法预测肾毒性。尽管基于QSAR的方法能在分子水平实现结构-毒性关系的定量分析，但其在草药毒性预测中的准确性和适用性受限于结构和毒理学数据的可用性与质量，以及许多成分表征不完整的问题。

网络毒理学是基于网络药理学生物医学研究的重要分支（Li等人，2019）。网络毒理学的主要研究目的是通过特定的毒性-靶点-成分-草药相互作用网络分析草药毒性。一般分析流程包括：识别草药的毒性成分、预测毒性成分以及毒性靶点的功能分析。Wu等人（2019a）结合四种机器学习方法与四种不同指纹集构建了16个单一分类器来预测潜在肝毒性成分，并通过系统药理学方法探索了草药的肝毒性机制。Yu等人（2025）在正清风痛宁诱导的肝损伤数据集中比较了十种机器学习模型，以寻找预测肝损伤最有效的模型。虽然网络毒理学能够系统阐明毒性机制并识别关键靶点，但这类方法无法直接量化草药水平毒性，且受限于不完整的草药-成分-靶点关联，导致预测存在偏差或不完整。

“四气五味”（又称“四气五味”）与归经共同构成了草药药理学理论框架的核心组成部分（Qiao等人，2022）。草药的“四气”分为五种类型：寒、热、

toxicity prediction is often simplified as a binary classification task, whereas it should inherently be a multi-label classification task, as a herb is often associated with different toxic effects. In our collected TCM toxicity dataset, more than half of the herbs (141 out of 252) are linked with two or more toxic labels, highlighting the complexity and necessity of multi-label learning based herb toxicity prediction. Fourth, deep learning models are often seen as “black boxes”, making them difficult to provide clear biological explanations for their toxicity prediction (Gao et al., 2025). This lack of interpretability hinders researchers’ understanding of toxicity mechanisms, and thus undermines the authenticity and clinical applicability of deep models.

Regulatory authorities and researchers have been emphasizing the importance of improving the safety and quality control of herb products (Jit̃areanu et al., 2022). Traditional methods for assessing herb toxicity, such as in vivo animal testing and in vitro cellular assays, are not only time-consuming and costly but also raise ethical concerns. Moreover, given the vast number of ingredients and their diverse action mechanisms, the comprehensive toxicity assessments remain a daunting task. With the advent of high-throughput technologies and the growing availability of large-scale biological and chemical datasets, computational approaches have gained attraction in herb toxicity prediction. These approaches include machine learning models, such as random forests, support vector machines, and deep learning frameworks, which can analyze complex, multidimensional data to predict toxicity outcomes (He et al., 2019a; Sun et al., 2019; Yang et al., 2021). By integrating data on chemical structures, pharmacological activities, and known toxicological profiles, these models offer a cost-effective and scalable solution for identifying potential toxicity of herbs and facilitating their safe usage (Jia et al., 2024; Wang et al., 2020; Wu et al., 2019a; Yu et al., 2025). However, existing computational approaches still encounter significant challenges arising from the heterogeneity and incompleteness of herb toxicity related data. Most current models are limited to a single type of information, such as molecular structure or pharmacological activity, and often lack the capability to effectively integrate diverse data sources that reflect the complex biological mechanisms of herbs. Furthermore, the absence of systematic frameworks capable of capturing multi-level interactions among herbs, ingredients, targets, and toxicities, restricts both the interpretability and authenticity of toxicity predictions. Given these, there is a critical need of authentic computational approaches that can systematically fuse multi-source information and address the challenges of data heterogeneity, sparsity, and interpretability in herb toxicity prediction.

In this paper, we construct a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies to solve the problem of dispersed and scarce herb toxicity data (see Section 4). To address the difficulties in modeling complex herbs and the lack of interpretability, we propose a heterogeneous network based approach (HerbToxNet) for herb toxicity prediction. HerbToxNet first constructs a herb-ingredient-target heterogeneous network using the pharmacological properties, toxicological characteristics, and associations between herbs, and then uses Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) to learn the representations of herbs. At the same time, it introduces contrastive learning with dynamic coefficient to pull closer the representations of herb pairs with shared toxic labels while pushing apart those with different toxic labels. Next, it uses a Multilayer Perceptron (MLP) on the herb representation to predict the initial toxic labels, and further introduces a weighted label fusion strategy that retrieves similar herbs and merges their toxic labels to refine the predicted labels.

Our contributions are summarized as below:

(i) We construct and share a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies scattered in different herb databases and published literature.

(ii) We propose a unified framework HerbToxNet to model multi-level relationships among herbs, ingredients, and targets via a heterogeneous network, which allows for transparent analysis of toxicity-related mechanisms and enhances interpretability.

(iii) HerbToxNet synergizes heterogeneous network modeling, contrastive learning, and weighted label fusion to remedy scarce toxicity data and achieve toxicity prediction in a sensible and authentic way. (iv) Extensive experiments show that HerbToxNet outperforms competitive methods, identifies potential toxicities of canonical herbs (96% confirmed), and can directly uncover toxicity-related ingredients and targets from an interpretable view.

The rest of the paper is organized as follows. Section 2 discusses the related work and Section 3 elaborates on the proposed HerbToxNet framework. Section 4 presents the experimental setup, results and analysis. Section 5 summarizes our work and outlines future efforts.

2. Related work

Existing computational methods for herb toxicity usually predict some main types of toxicity, such as hepatotoxicity, nephrotoxicity, cardiotoxicity, etc., and model them as binary classification tasks (whether the interested toxicity is present or not). Based on the characteristics of herb toxicity, a series of theories and methods for herb toxicity prediction were developed, which can be divided into three categories: herb toxicity prediction based on quantitative structure-activity relationship (QSAR), network toxicology, herb property theory.

In computational toxicology, the virtual structure of a compound is often used to analyze, simulate, visualize, or predict the toxicity of a chemical by computational methods (Raies & Bajic, 2016). QSAR can quantitatively capture the relationship between molecular structure and biological effects, it empowers the computational prediction of compound toxicity and has been widely applied in the toxicity assessment of chemical drugs (Cherkasov et al., 2014). Therefore, QSAR-based methods can be adapted to computationally predict the toxicity of individual ingredients (alike drugs) within each herb, providing a basis for estimating the overall toxicity of herbs (Gao et al., 2019). He et al. (2019a) collected hepatotoxic herbs and ingredients, analyzed the structure of these ingredients and found that alkaloids and terpenoids were the main hepatotoxic ingredients. Yang et al. (2021) used QSAR to evaluate the potential toxicity of ingredients in Cassia seed, and clarified the relationship between structural characteristics and cytotoxicity by linear model, and found that three ingredients in Cassia seed had hepatotoxicity and eight ingredients had nephrotoxicity. Sun et al. (2019) used herb nephrotoxicity data to construct a QSAR model, and used ANN and SVM algorithms to predict nephrotoxicity. While QSAR-based approaches enable quantitative analysis of structure-toxicity relationships at the molecular level, their accuracy and applicability in herbal toxicity prediction are limited by the availability and quality of both structural and toxicological data, as well as the incomplete characterization of many ingredients.

Network toxicology, is an important branch for studying biomedicine based on network pharmacology (Li et al., 2019). The main research purpose of network toxicology is to analyze herb toxicity through a specific toxicity-target-ingredient-herb interaction network. The general analysis process includes: identification of toxic ingredients of herb, prediction of toxic ingredients and functional analysis of toxic targets. Wu et al. (2019a) combined four machine learning methods with four different fingerprint sets to construct 16 single classifiers to predict potential hepatotoxic ingredients, and explored the mechanism of hepatotoxicity of herbs through systematic pharmacological methods. Yu et al. (2025) compared ten machine learning models in the Zhengqing Fengtongning-induced liver injury dataset to find the most effective model for predicting liver injury. While network toxicology can systematically elucidate toxic mechanisms and identify key targets, this kind of methods cannot directly quantify herb-level toxicity and are limited by incomplete herb-ingredient-target associations, suffering biased or incomplete predictions.

The “Four Properties and Five Flavors” (also known as the “Four Qi and Five Tastes”) and meridian tropism constitute critical components of the theoretical framework of herb pharmacology (Qiao et al., 2022). The “Four Properties” of herb are categorized into five types: cold, hot,

温、凉和平。而“五味”则分为七种类型：酸、苦、甘、辛、咸、淡和涩。归经是指将草药归类至十二经络：肺、大肠、胃、脾、心、小肠、膀胱、肾、心包、三焦、胆和肝。Wang 等人（2020）建立了一个随机游走网络，该网络能够在传统人工神经网络的基础上有效解读处方信息，以预测草药处方的副作用。Jia 等人（2024）通过将草药毒性与电子电离质谱数据的高级分析描述符相关联，结合可解释的机器学习模型进行了毒性预测。此外，一些研究表明，草药的功效可用于分析其毒性（Yang 等人，2015；Zhu 等人，2025）。虽然药性理论提供了植根于传统知识的描述性见解，但其定性和主观的特性，与用于化学药物的标准化表示不同，导致数据模糊不清，即使结合先进的机器学习方法，其预测性能也有限。

为解决上述挑战，我们构建了一个异质图来建模草药、成分和靶点之间的复杂关系。通过利用异质图注意力网络、动态系数对比学习和加权标签融合，我们的HerbToxNet能够处理不完整数据并融合多层次特征，从而有效预测草药的毒性标签。

3. 所提出的 方法

HerbToxNet的框架如图1所示。该框架包含四个主要模块：(i) 异质图构建：基于草药药理学特性、功效和关联构建草药-成分-靶点异质图；(ii) 草药表示学习：采用异质图注意力网络（Wang等人，2019年）结合选定元路径来提取信息丰富的草药表征；(iii) 带动态系数的对比学习：通过拉近具有共享

毒性标签的草药距离并推远标签不同的草药，优化学习到的表征；(iv) 基于加权标签融合的毒性预测：将多层感知器的预测结果与目标草药相似草药的毒性标签基于其优化后的表征进行融合。这些模块按序组织，使HerbToxNet能够有效整合异质信息、学习鲁棒表征并实现毒性预测。

3.1. 异质图 构建

草药药理特性和功效数据包含用于毒性预测的有用信息，但由于这些数据基于传统文献和经验知识收集，往往存在不完整或不精确的问题。因此，基于草药药理特性和功效构建的现有方法难以充分发挥其潜力。事实上，草药具有多种具有协同效应的成分，使其毒性变得复杂，仅凭药理特性数据难以描述和捕捉（Caesar & Cech, 2019; Flora et al., 2013）。同样，网络毒理学也面临挑战，因为成分的复杂性使得现有方法难以准确预测其毒性效应。此外，这些模型通常缺乏可解释性，阻碍了对生物学机制的理解，并降低了其在实际应用中的可靠性（He 等人，2019a,b）。

为解决上述问题，我们将异质草药-成分-靶点图定义为 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ ，通过联合建模成分和靶点节点及其相互作用，来编码草药的多成分和多靶点特性。节点集 $\mathcal{V} = \mathcal{V}_h \cup \mathcal{V}_c \cup \mathcal{V}_t$ 是草药、成分和靶点节点的并集，边集 $\mathcal{E} = \mathcal{E}_{hc} \cup \mathcal{E}_{ht} \cup \mathcal{E}_{ct} \cup \mathcal{E}_{hh}$ 是草药-成分、草药-靶点、成分-靶点和草药-草药边的并集。我们的目标是将每味草药的药理学特性和功效都整合到其节点表示中，从而将领域知识与异质图的结构信息相结合。第*i*味草药的药理学特性和功效分别表示为 $\mathbf{p}_i \in \{0, 1\}^{L_p}$ 和 $\mathbf{e}_i \in \{0, 1\}^{L_e}$ ，

warm, cool, and neutral. The “Five Flavors” are classified into seven types: sour, bitter, sweet, pungent, salty, bland, and astringent. Meridian tropism refers to classifying herbs into 12 channels: lung, large intestine, stomach, spleen, heart, small intestine, bladder, kidney, pericardium, triple burner, gallbladder, and liver. Wang et al. (2020) established a random walk network that can effectively interpret prescription information on the basis of traditional ANN to predict side effects of herb prescriptions. Jia et al. (2024) made toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. In addition, some studies have shown that the efficacy of herbs can be used to analyze herb toxicity (Yang et al., 2015; Zhu et al., 2025). While herb property theory offers descriptive insights rooted in traditional knowledge, its qualitative and subjective nature, distinct from the standardized representations used for chemical drugs, results in ambiguous data and limited predictive performance even when combined with advanced machine learning methods.

To address the above challenges, we construct a heterogeneous graph to model the complex relationships between herbs, ingredient and targets. By leveraging Heterogeneous Graph Attention Networks, contrastive learning with dynamic coefficient, and weighted label fusion, our HerbToxNet can handle incomplete data and fuse multi-level features, thereby effectively predicting the toxic labels of herbs.

3. The proposed methodology

The framework of HerbToxNet is illustrated in Fig. 1. HerbToxNet comprises four main modules: (i) Heterogeneous graph construction builds a herb-ingredient-target heterogeneous graph based on herb pharmacological properties, efficacy, associations; (ii) Herb representation learning employs a Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) with selected meta-paths to extract informative herb representations; (iii) Contrastive learning with dynamic coefficient optimizes the learned representations by pulling together herbs with shared

toxic labels and pushing apart those with dissimilar labels; and (iv) Toxicity prediction based on weighted label fusion merges the predictions from an MLP and toxic labels from similar herbs of the target herb based on their optimized representations. These modules are organized sequentially, enabling HerbToxNet to effectively integrate heterogeneous information, learn robust representations, and achieve toxicity prediction.

3.1. Heterogeneous graph construction

Herb pharmacological property and efficacy data contain useful information for toxicity prediction, but they are often incomplete or imprecise, as they are collected based on traditional literature and empirical knowledge. As such, existing methods built on herb pharmacological property and efficacy often struggle to exploit their potential. In fact, a herb has multiple ingredients with synergistic effects, making its toxicity complex and difficult to describe and capture with pharmacological property data alone (Caesar & Cech, 2019; Flora et al., 2013). Similarly, network toxicology faces challenges, as the complexity of ingredients makes it difficult for existing methods to accurately predict their toxic effects. Furthermore, these models often lack interpretability, hindering the understanding of biological mechanisms and reducing their reliability in practical applications (He et al., 2019a,b).

To address the above issues, we define the heterogeneous herb-ingredient-target graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, which encode the multi-ingredient and multi-target nature of herbs by jointly modeling ingredient and target nodes, as well as their interactions. The node set $\mathcal{V} = \mathcal{V}_h \cup \mathcal{V}_c \cup \mathcal{V}_t$ is the union of herb, ingredient, and target nodes, and the edge set $\mathcal{E} = \mathcal{E}_{hc} \cup \mathcal{E}_{ht} \cup \mathcal{E}_{ct} \cup \mathcal{E}_{hh}$ is the union of herb-ingredient, herb-target, ingredient-target, and herb-herb edges. We aim to integrate both the pharmacological properties and efficacy of each herb into its node representation, thereby combining domain knowledge with structural information of the heterogeneous graph. The pharmacological properties and efficacy of *i*th herb are denoted as $\mathbf{p}_i \in \{0, 1\}^{L_p}$ and $\mathbf{e}_i \in \{0, 1\}^{L_e}$,

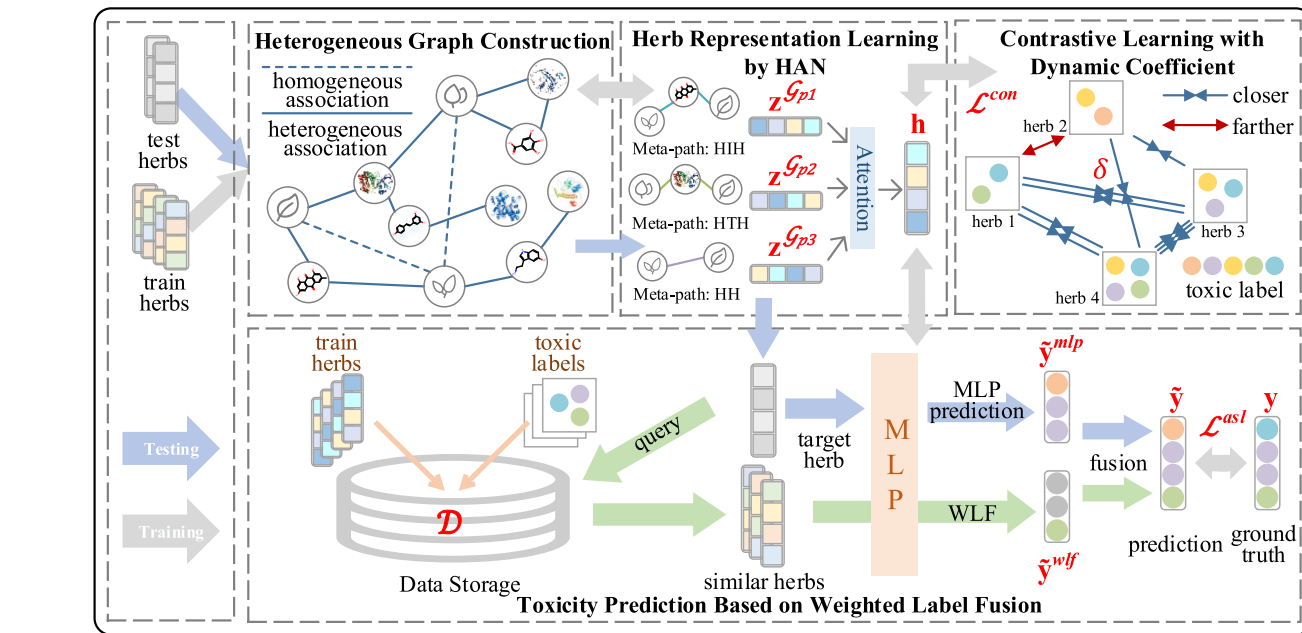


Fig. 1. The overview of HerbToxNet framework. HerbToxNet first constructs a herb-ingredient-target heterogeneous graph and selects three types of meta-paths: herb-ingredient-herb (HIH), herb-target-herb (HTH) and herb-herb (HH) to build the meta-path matrix. Then it utilizes the attention module of heterogeneous graph neural network to synthesize features ($\mathbf{z}^{Gp_1}$, $\mathbf{z}^{Gp_2}$, $\mathbf{z}^{Gp_3}$) under different meta-paths to obtain the representation ($\mathbf{h}$) of the herb. Next, it introduces a contrastive learning loss ($\mathcal{L}^{con}$) with a dynamic coefficient δ to pull together the representations of herbs with shared toxic labels, while pushing apart those without shared labels thereby optimizing herb representations. After that, it predicts the toxic labels in two ways: an MLP classifier to generate $\tilde{\mathbf{y}}^{mlp}$ and a weighted label fusion strategy to aggregate labels $\tilde{\mathbf{y}}^{wlf}$ from similar herbs in the training set $\mathcal{D}$, and then merges $\tilde{\mathbf{y}}^{mlp}$ and $\tilde{\mathbf{y}}^{wlf}$ into $\tilde{\mathbf{y}}$. Finally, HerbToxNet defines asymmetric classification loss ($\mathcal{L}^{asl}$) between $\tilde{\mathbf{y}}$ and the ground-truth labels $\mathbf{y}$ to optimize the overall framework.

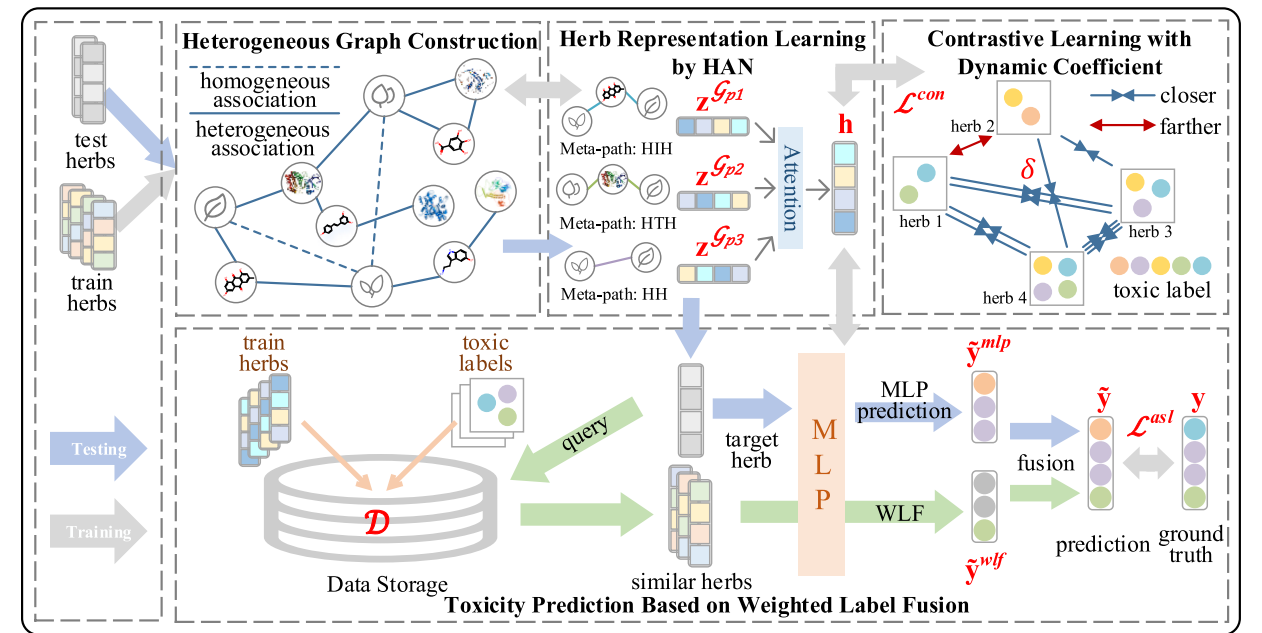


Fig. 1. The overview of HerbToxNet framework. HerbToxNet first constructs a herb-ingredient-target heterogeneous graph, and selects three types of meta-paths: herb-ingredient-herb (HIH), herb-target-herb (HTH), and herb-herb (HH) to build the meta-path matrix. Then, it utilizes the attention module of heterogeneous graph neural network to synthesize features ($\mathbf{z}^{Gp_1}$, $\mathbf{z}^{Gp_2}$, $\mathbf{z}^{Gp_3}$) under different meta-paths to obtain the representation ($\mathbf{h}$) of the herb. Next, it introduces a contrastive learning loss ($\mathcal{L}^{con}$) with a dynamic coefficient δ to pull together the representations of herbs with shared toxic labels, while pushing apart those without shared labels, thereby optimizing herb representations. After that, it predicts the toxic labels in two ways: an MLP classifier to generate $\tilde{\mathbf{y}}^{mlp}$, and a weighted label fusion strategy to aggregate labels $\tilde{\mathbf{y}}^{wlf}$ from similar herbs in the training set $\mathcal{D}$, and then merges $\tilde{\mathbf{y}}^{mlp}$ and $\tilde{\mathbf{y}}^{wlf}$ into $\tilde{\mathbf{y}}$. Finally, HerbToxNet defines asymmetric classification loss ($\mathcal{L}^{asl}$) between $\tilde{\mathbf{y}}$ and the ground-truth labels $\mathbf{y}$ to optimize the overall framework.

其中 L_p 和 L_e 分别表示药理学特性和功效的维度。若第味草药具有第 j 种药理学特性, 则 $p_{ij} = 0$ 为1, 否则为0。若第味草药包含第种功效, 则 $e_{ij} = 0$ 为1, 否则为0。我们使用草药的药理学特性和功效作为其初始节点特征。对于第味草药, 其特征表示 $\mathbf{f}_i$ 是通过将 $\mathbf{p}_i$ 和 $\mathbf{e}_i$ 的拼接应用变换得到的:

$$\mathbf{f}_i = \text{ReLU}(\mathbf{W}_h(\mathbf{p}_i || \mathbf{e}_i) + \mathbf{b}_h) \quad (1)$$

其中 $\mathbf{W}_h$ 和 $\mathbf{b}_h$ 分别表示可训练权重矩阵和偏置向量。 $||$ 表示拼接操作, $\text{ReLU}(\cdot)$ 是ReLU激活函数。由于我们的目标是草药毒性预测, 且异质图将基于预定义的元路径使用HAN进行表示, 因此无需为成分和靶点节点分配特定的初始特征。

异质图 $\mathcal{G}$ 包含三种节点类型: 草药、成分和靶点, 通过三种异质关系 (草药-成分、草药-靶点和成分-靶点) 以及一种同质关系 (草药-草药) 连接。成分间或靶点间的同质关系需要显式的相似度计算或人工标注, 这会引入冗余和繁琐工作, 因此本研究未使用这些关系。该图结构为建模草药的结构和属性信息提供了统一工具, 有助于毒性预测。

3.2. 基于HAN的 草药 表示 学习

在异质图 $\mathcal{G}$ 上给定一条元路径 p , 该元路径采用 $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \dots \xrightarrow{r_l} v_{l+1}$ 的形式, 表示节点类型 v_{τ_1} 和 $v_{\tau_{l+1}}$ 通过复合关系 $r = r_1 \circ r_2 \circ \dots \circ r_l$ 连接, 其中 $\circ$ 是关系上的复合运算符, $v_i \in \mathcal{V}$ 和 $r_i \in \mathcal{E}$ 。换言之, 关系 r 由那些 $r_1, r_2, \dots, r_l$ 的关系组合而成。因此, 元路径能够捕捉异质图中的特定语义信息, 不同的元路径代表不同的语义内容。

例如, 草药-成分-草药 (HIH) 元路径表明两种草药含有相同成分, 这意味着这两种草药可能具有相似的功能和靶点关联。我们通过将异质图中草药与成分的关联矩阵与其转置相乘, 基于HIH元路径构建矩阵。类似地, 草药-靶点-草药 (HTH) 元路径表明两种草药与同一靶点相连。草药-草药 (HH) 元路径则表示两种草药之间的相似性。HTH和HH矩阵的构建方法与HIH矩阵类似。

在此, 我们将异质图 $\mathcal{G}$ 分解为多个子图, 每个子图 $\mathcal{G}_p$ 由通过元路径 p 连接的所有相邻节点对组成, 因此被指定为基于元路径的子图。具体而言, 对于以草药节点为起点或终点的元路径集合, 我们将其表示为 $\mathcal{P} = \{p_1, p_2, \dots, p_M\}$, 其中 $M = |\mathcal{P}|$ 是所考虑的元路径数量。

异质图注意力网络 (HAN) (Wang等人, 2019年) 是一种用于异质图上图表示学习的有效工具, 能够通过分层注意力机制为不同邻近样本和元路径分配不同权重。在此背景下, 我们利用HAN来学习每个元路径诱导子图中的节点表示。HAN自适应地聚合每个子图中节点的邻近样本信息, 以整合节点拓扑和特征。具体而言, 对于给定节点及其关联子图, HAN首先采用节点级自注意力机制来学习其邻近节点的重要性 (注意力分数)。然后, 根据邻近样本的注意力分数自适应地聚合其特征表示。最后, 使用语义级注意力机制来组合来自不同元路径的表示。

对于第 k 个基于元路径的子图 $\mathcal{G}_{p_k}$, 节点对 (i, j) 的重要性 $e_{ij}^{G_{p_k}}$ 可表述如下:

$$e_{ij}^{G_{p_k}} = \text{LeakyReLU}(\mathbf{a}_{G_{p_k}}^T \cdot [\mathbf{f}_i^{G_{p_k}} || \mathbf{f}_j^{G_{p_k}}]) \quad (2)$$

其中 $\mathbf{a}_{G_{p_k}}$ 是第 k 条元路径的注意力参数。为使不同元路径的注意力系数具有可比性, 我们采用softmax函数对 $e_{ij}^{G_{p_k}}$ 进行归一化:

$$e_{ij}^{G_{p_k}'} = \frac{\exp(e_{ij}^{G_{p_k}})}{\sum_{j' \in \mathcal{V}_{hi}^{G_{p_k}}} \exp(e_{ij'}^{G_{p_k}})} \quad (3)$$

其中 $\mathcal{V}_{hi}^{G_{p_k}}$ 表示节点 i 在第 k 条元路径中的邻居节点集合。接下来, 我们使用ReLU激活函数聚合所有元路径实例的特征向量, 从而得到基于第 k 条元路径的节点 i 表示:

$$\mathbf{f}_i^{G_{p_k}} = \text{ReLU} \left(\sum_{j \in \mathcal{V}_{hi}^{G_{p_k}}} e_{ij}^{G_{p_k}'} \cdot \mathbf{f}_j^{G_{p_k}} \right) \quad (4)$$

在本研究中, 我们引入了多头注意力机制以增强注意力系数学习过程的稳定性, 并减轻单个注意力头的影响。具体而言, 我们独立重复注意力机制 K 次, 并将每个注意力头学习到的向量表示进行拼接。因此, 节点 i 在第 k 条元路径下的表示 $\mathbf{z}_i^{G_{p_k}}$ 由下式给出:

$$\mathbf{z}_i^{G_{p_k}} = ||_{j=1}^K \mathbf{f}_{ij}^{G_{p_k}} \quad (5)$$

通常, 异质图中的每个节点包含多种类型的语义信息, 而单一元路径下的节点特征聚合只能反映节点信息的一个方面。为学习更全面的节点嵌入, 必须整合多条元路径所代表的语义信息。聚合不同元路径嵌入的一种简单方法是取平均值。然而, 由于元路径对异质图中节点 i 的重要性各不相同, 我们采用注意力机制为每个元路径模式分配权重, 然后进行聚合。这确保了具有最关键字信息的嵌入被赋予最高的重要性。更具体地说, 我们基于每个元路径的子图学习其重要性, 并得到 $|\mathcal{P}|$ 组节点嵌入, 其中每组表示对应元路径下所有节点的加权聚合:

$$w_{G_{p_k}} = \frac{1}{|\mathcal{V}_h|} \sum_{i \in \mathcal{V}_h} \mathbf{q}^T \cdot \tanh(\mathbf{W}_k \cdot \mathbf{z}_i^{G_{p_k}} + \mathbf{b}_k) \quad (6)$$

其中 $\mathbf{W}_k$ 、 $\mathbf{b}_k$ 和 $\mathbf{q}_k$ 分别表示与第 k 个元路径相关联的权重矩阵、偏置向量和语义级注意力向量。 $\tanh(\cdot)$ 代表tanh激活函数。在获得特定节点不同元路径的注意力分数后, 我们使用softmax函数对其进行归一化处理, 从而实现对不同元路径间语义信息的有意义比较。

$$w_{G_{p_k}}' = \frac{\exp(w_{G_{p_k}})}{\sum_{j=1}^{|\mathcal{P}|} \exp(w_{G_{p_j}})} \quad (7)$$

最后, 我们基于各自的注意力分数, 整合来自不同元路径的节点嵌入向量。第 i 个草药节点 $\mathbf{h}_i$ 嵌入向量 $\mathbf{h}_i$ 如下所示:

$$\mathbf{h}_i = \sum_{k=1}^{|\mathcal{P}|} w_{G_{p_k}}' \cdot \mathbf{z}_i^{G_{p_k}} \quad (8)$$

respectively, where L_p and L_e represent the dimension of pharmacological properties and efficacy. $p_{ij} = 1$ if i th herb possesses the j th pharmacological properties, and $p_{ij} = 0$ otherwise. $e_{ij} = 1$ if i th herb contains the j th efficacy, and $e_{ij} = 0$ otherwise. We use the pharmacological properties and efficacy of herbs as their initial node features. For the i th herb, its feature representation $\mathbf{f}_i$ is obtained by applying a transformation to the concatenation of $\mathbf{p}_i$ and $\mathbf{e}_i$ as:

$$\mathbf{f}_i = \text{ReLU}(\mathbf{W}_h(\mathbf{p}_i || \mathbf{e}_i) + \mathbf{b}_h) \quad (1)$$

where $\mathbf{W}_h$ and $\mathbf{b}_h$ denote the trainable weight matrix and bias vector, respectively. $||$ represents the concatenation operation, and $\text{ReLU}(\cdot)$ is the ReLU activation function. Since our objective is herb toxicity prediction, and the heterogeneous graph will be represented using HAN based on predefined meta-paths. Therefore, there is no need to assign specific initial features to the ingredient and target nodes.

The heterogeneous graph $\mathcal{G}$ consists of three node types: herb, ingredient, and target, which are connected by three types of heterogeneous relations (herb-ingredient, herb-target, and ingredient-target), as well as one type of homogeneous relation (herb-herb). Homogeneous relations between ingredients or targets require explicit similarity computations or manual curation, which introduce redundancy and tedious efforts, so these relations are not used in our work. This graph structure provides a unified tool for modeling both the structure and attribute information of herbs, facilitating toxicity prediction.

3.2. Herb representation learning by HAN

Give a meta-path p on the heterogeneous graph $\mathcal{G}$, with the meta-path taking the form of $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \dots \xrightarrow{r_l} v_{l+1}$, denoting that the node types v_1 and v_{l+1} are connected by a composite relationship $r = r_1 \circ r_2 \circ \dots \circ r_l$, where $\circ$ is the composite operator on the relationship, $v_i \in \mathcal{V}$ and $r_i \in \mathcal{E}$. In other words, the relationship r is composed of l relationships of those $r_1, r_2, \dots, r_l$. Thus, meta-path can capture specific semantic information within a heterogeneous graph, with different meta-paths representing different semantic contents.

For example, the meta-path of Herb-Ingredient-Herb (HIH) suggests that both herbs contain the same ingredient, implying that these two herbs may possess similar functionalities and target associations. We construct the matrix based on the HIH meta-path by multiplying the association matrix of herbs and ingredients in the heterogeneous graph with its transpose. Similarly, meta-path Herb-Target-Herb (HTH) suggests that both herbs link with the same target. The meta-path Herb-Herb (HH) indicates the similarity between two herbs. The construction methods for the HTH and HH matrices are similar to that of the HIH matrix.

Here, we decompose the heterogeneous graph $\mathcal{G}$ into multiple subgraphs, each subgraph $\mathcal{G}_p$ is composed of all neighbor pairs connected by the meta-path p , thereby designated as meta-path based subgraphs. Specifically, for the set of meta-paths that start or end with herb nodes, we denote it as $\mathcal{P} = \{p_1, p_2, \dots, p_M\}$, where $M = |\mathcal{P}|$ is the number of meta-paths considered.

The Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) is an effective tool for graph representation learning on heterogeneous graphs, capable of assigning different weights to different neighbors and meta-paths via a hierarchical attention mechanism. In this context, we utilize HAN to learn node representations within each meta-path induced subgraph. HAN adaptively aggregates the neighbor information of nodes in each subgraph to integrate both node topology and features. Specifically, for a given node and its associated subgraph, HAN first employs a node-level self-attention mechanism to learn the importance (attention scores) of its neighboring nodes. Then, it adaptively aggregates the feature representations of neighbors based on their attention scores. Finally, a semantic-level attention mechanism is used to combine representations from different meta-paths.

For the k th meta-path based subgraph $\mathcal{G}_{p_k}$, the importance $e_{ij}^{G_{p_k}}$ of node pair (i, j) can be formulated as follows:

$$e_{ij}^{G_{p_k}} = \text{LeakyReLU}(\mathbf{a}_{G_{p_k}}^T \cdot [\mathbf{f}_i^{G_{p_k}} || \mathbf{f}_j^{G_{p_k}}]) \quad (2)$$

where $\mathbf{a}_{G_{p_k}}$ is the attention parameter of the k th meta-path. To make the attention coefficients of different meta-paths comparable, we use the softmax function to normalize $e_{ij}^{G_{p_k}}$:

$$e_{ij}^{G_{p_k}'} = \frac{\exp(e_{ij}^{G_{p_k}})}{\sum_{j' \in \mathcal{V}_{hi}^{G_{p_k}}} \exp(e_{ij'}^{G_{p_k}})} \quad (3)$$

where $\mathcal{V}_{hi}^{G_{p_k}}$ represents the set of neighbor nodes of node i in the k th meta-path. Next, we aggregate the feature vectors of all meta-path instances using the ReLU activation function to derive the representation of node i based on the k th meta-path:

$$\mathbf{f}_i^{G_{p_k}} = \text{ReLU} \left(\sum_{j \in \mathcal{V}_{hi}^{G_{p_k}}} e_{ij}^{G_{p_k}'} \cdot \mathbf{f}_j^{G_{p_k}} \right) \quad (4)$$

In this study, we introduce a multi-head attention mechanism to enhance the stability of the attention coefficient learning process and mitigate the impact of individual attention heads. Specifically, we independently repeat the attention mechanism K times and concatenate the vector representations learned by each attention head. Thus, the representation $\mathbf{z}_i^{G_{p_k}}$ of node i under the k th meta-path is given by:

$$\mathbf{z}_i^{G_{p_k}} = ||_{j=1}^K \mathbf{f}_{ij}^{G_{p_k}} \quad (5)$$

Generally, each node in the heterogeneous graph contains multiple types of semantic information, and the aggregation of nodes features under a single meta-path can only reflect one aspect of the node's information. To learn more comprehensive node embeddings, it is essential to integrate the semantic information represented by multiple meta-paths. A simple method for aggregating embeddings from different meta-paths is to take the average. However, due to the varying importance of meta-paths to node i in the heterogeneous graph, we adopt an attention mechanism to assign weights to each meta-path pattern and then perform aggregation. This ensures that the embeddings with the most crucial semantic information are given the most significance. More specifically, we learn the importance of each meta-path based subgraph and obtain $|\mathcal{P}|$ sets of node embeddings, where each set represents the weighted aggregation of all nodes under the corresponding meta-path:

$$w_{G_{p_k}} = \frac{1}{|\mathcal{V}_h|} \sum_{i \in \mathcal{V}_h} \mathbf{q}^T \cdot \tanh(\mathbf{W}_k \cdot \mathbf{z}_i^{G_{p_k}} + \mathbf{b}_k) \quad (6)$$

where $\mathbf{W}_k$, $\mathbf{b}_k$, and $\mathbf{q}_k$ denote the weight matrix, bias vector, and semantic-level attention vector associated with the k th meta-path. $\tanh(\cdot)$ represents the tanh activation function. After obtaining the attention scores of different meta-paths for a specific node, we normalize them using the softmax function, allowing for meaningful comparisons of the semantic information across different meta-paths.

$$w_{G_{p_k}}' = \frac{\exp(w_{G_{p_k}})}{\sum_{j=1}^{|\mathcal{P}|} \exp(w_{G_{p_j}})} \quad (7)$$

Finally, we integrate the node embeddings from different meta-paths based on their respective attention scores. The i th herb node embedding $\mathbf{h}_i$ is as follows:

$$\mathbf{h}_i = \sum_{k=1}^{|\mathcal{P}|} w_{G_{p_k}}' \cdot \mathbf{z}_i^{G_{p_k}} \quad (8)$$

3.3. 基于 动态系数 的 对比 学习

上述草药表征可能无法有效捕捉草药间与毒性相关的相似性和差异性，特别是在每个草药可能拥有多个毒性标签的场景下。因此，我们提出采用动态系数策略的对比学习来更新草药表征。基于典型对比学习的方法旨在最小化同类实例间的距离，同时最大化不同类实例间的距离（Chen et al., 2020; Hadsell et al., 2006; Khosla et al., 2020）。然而，在多标签草药毒性预测中，两个草药通常仅共享部分毒性标签，且每个草药都具有独特的毒性标签。这种多标签现象比单标签设置更为复杂，使得难以简单地将两个草药视为“正样本对”或“负样本对”。

为捕捉具有多重毒性标签的草药间复杂关系，我们基于毒性标签语义相似度定义了动态系数。与单标签设置中的二值指示器不同，该系数通过数值化量化草药对间的标签重叠程度，为模型提供了灵活处理多标签毒性预测任务中固有复杂关系的能力。具体而言，我们通过以下方式衡量两种草药的特征相似度：

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{\|\mathbf{h}_i\|_2 \|\mathbf{h}_j\|_2} \quad (9)$$

其中 $\cos(\dots)$ 表示 余弦 相似度， $\mathbf{h}_i$ 和 $\mathbf{h}_j$ 分别是 第 i 个 与 第 j 个 草药的 学习所得 嵌入向量， $\|\cdot\|_2$ 代表 L_2 范数。

为 进一步 整合 毒性 标签 关联性，我们 将 标签 语义 相似度 β_{ij} 定义 为 两种草药间 共享毒性标签 的 比例：

$$\beta_{ij} = \frac{\mathbf{y}_i^T \cdot \mathbf{y}_j}{\mathbf{y}_i^T \cdot \mathbf{y}_i} \quad (10)$$

其中 $\mathbf{y}_i \in \{0, 1\}^L$ 和 $\mathbf{y}_j \in \{0, 1\}^L$ 分别表示第 i 个和第 j 个草药的多热毒性标签向量， L 为毒性标签总数。分子 $\mathbf{y}_i^T \cdot \mathbf{y}_j$ 统计两种草药共享的毒性标签数量，分母 $\mathbf{y}_i^T \cdot \mathbf{y}_i$ 则表示第 i 个草药的毒性标签总数。这种基于比率的语义相似度使模型不仅能区分单标签设置中完全重合或互斥的标签集，还能量化多标签场景特有的部分相似性。

在后续实验中，我们发现 ρ 可能不足以作为对比学习的动态系数，因为它无法提供两个草药之间相关性的稳健度量。特别是 ρ 会给毒性标签相似度较低的草药样本分配过大的系数。这个问题会降低草药毒性预测的性能。因此，我们引入可调超参数 δ 来调整 ρ ，以获得更合适的动态系数 δ ：

$$\delta_{ij} = (\beta_{ij})^t \quad (11)$$

具有动态系数 δ_{ij} 的 一对 草药 的对比学习损失定义为：

$$\mathcal{L}_{ij}^{con} = -\delta_{ij} \log \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau)}{\sum_{k=1, k \neq i}^N \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau)} \quad (12)$$

其中 τ 是用于缩放相似度得分的温度超参数， N 是批次中草药的总数。该公式使模型能够强调具有更高毒性相似度的相似草药对，同时有效学习具有多个相关毒性标签的草药表示。此公式明确利用了草药的多标签毒性特性，因为每个草药对损失的贡献程度由其共享毒性标签的范围来缩放。

对于给定的草药对 $(\mathbf{h}_i, \mathbf{h}_j)$ ，较高的语义相似度 β_{ij} 会产生较大的动态系数 δ_{ij} ，这继而导致较大的损失 $\mathcal{L}_{ij}^{con}$ 。因此，特征相似度 $\cos(\mathbf{h}_i, \mathbf{h}_j)$ 被优化朝向更高值。相反，如果两个样本不共享任何标签（ $\beta_{ij} = \delta_{ij} = 0$ ），则值 $\mathcal{L}_{ij}^{con}$ 变为零，它们的余弦相似度 $\cos(\mathbf{h}_i, \mathbf{h}_j)$ 被优化至较低值。具有动态系数的对比学习损失针对 N 草药的计算公式为：

$$\mathcal{L}^{con} = \sum_{i=1}^N \sum_{j=1, j \neq i}^N \mathcal{L}_{ij}^{con} \quad (13)$$

3.4. 基于 加权 标签 融合的草药毒性预测

对于 第 i 味草药，其 表征向量 $\mathbf{h}_i$ 通过 一个 三层 多层感知器，再经由 sigmoid 激活 函数处理，最终计算出 该草药 对应所有毒性标签的 概率分布向量。

$$\tilde{\mathbf{y}}_i^{mlp} = \text{sigmoid}(\text{MLP}(\mathbf{h}_i)) \quad (14)$$

其中 $\text{MLP}(\cdot)$ 表示一个三层MLP。 $\tilde{\mathbf{y}}_i^{mlp} \in \{0, 1\}^L$ 表示第 i 个草药与毒性标签关联的预测概率向量。

然而，上述多层感知器可能无法充分利用训练数据中编码的宝贵知识。为了进一步提升毒性预测性能，我们引入了加权标签融合策略。对于每个目标草药，我们从训练集中检索具有相似表示的草药。通过对比学习，这些邻近样本更可能与目标草药共享有意义的毒性模式。HerbToxNet以加权方式聚合邻域草药的毒性标签，这补充了多层感知器的输出，有助于实现更准确的毒性预测。

令 $\mathcal{D} = \{(\mathbf{h}_j, \mathbf{y}_j)\}_{j=1}^N$ 表示包含 N 种草药的训练集。给定第 i 个目标草药及其表示 $\mathbf{h}_i$ ，我们通过计算 $\mathbf{h}_i$ 与 $\mathcal{D}$ 中所有 $\mathbf{h}_j$ 之间的余弦相似度，从训练集中检索其相关邻近样本。相似度超过预设阈值 σ 的邻近样本将被选入邻近集合：

$$\mathcal{N}_\sigma = \{(\mathbf{h}_j, \mathbf{y}_j) \mid \cos(\mathbf{h}_i, \mathbf{h}_j) \geq \sigma\} \quad (15)$$

与 使用 K-最近邻 不同，我们 基于阈值 的策略 自适应地 选择 那些 在 表示空间 中 与 目标 草药 接近 的 邻近样本。

基于 加权标签融合 的 预测 结果 针对 第 i 味 草药 通过 聚合 所选 邻近样本 的 标签向量 获得，这些向量 根据 其与 目标 草药 的 相似度 进行 加权：

$$\tilde{\mathbf{y}}_i^{wlf} = \sum_{(\mathbf{h}_j, \mathbf{y}_j) \in \mathcal{N}_\sigma} \mu_j \mathbf{y}_j \quad (16)$$

其中 权重 μ_j 定义 为：

$$\mu_j = \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau')}{\sum_{(\mathbf{h}_k, \mathbf{y}_k) \in \mathcal{N}_\sigma} \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau')} \quad (17)$$

其中 τ' 为控制权重分布锐度的温度超参数。通过这种方式，与第 i 味草药相似度更高的邻近样本被赋予更大权重，这些样本对基于加权标签融合的预测贡献显著优于其他邻近样本。

为此，我们将 基于多层感知器的 预测 $\tilde{\mathbf{y}}_i^{mlp}$ 与 基于加权标签融合的 预测 $\tilde{\mathbf{y}}_i^{wlf}$ 结合 为：

$$\tilde{\mathbf{y}}_i = \alpha \tilde{\mathbf{y}}_i^{wlf} + (1 - \alpha) \tilde{\mathbf{y}}_i^{mlp} \quad (18)$$

其中 标量参数 α 用于平衡 两种预测来源。

我们采用非对称损失（ASL）（Ridnik等人，2021）作为分类损失，以解决多标签草药毒性预测中不平衡的特性。ASL通过为正样本和负样本分配不同的惩罚，有效缓解了标签不平衡的影响，从而增强模型对罕见但关键毒性标签的关注。该损失

3.3. Contrastive learning with dynamic coefficient

The above herb representations may not effectively capture toxicity-related similarities and differences among herbs, especially under setting where each herb can possess multiple toxic labels. Therefore, we propose a contrastive learning with dynamic coefficient strategy to update the herb representations. Canonical contrastive learning based methods aim to minimize the distance between instances of the same class while maximizing that between instances of different ones (Chen et al., 2020; Hadsell et al., 2006; Khosla et al., 2020). However, in multi-label herb toxicity prediction, two herbs often share only a subset of toxic labels, and each herb posses unique toxic labels. This multi-label phenomenon imposes greater complexity than the single-label setting, making it hard to simply treat two herbs as either “positive” or “negative” pairs.

To capture the intricate relationships between herbs with multiple toxic labels, we define a dynamic coefficient based on toxic label semantic similarity. Unlike binary-valued indicators in single-label settings, this coefficient provides a flexible way to numerically quantify the degree of overlapped labels between herb pairs, allowing the model to account for the complex relationships inherent in multi-label toxicity prediction tasks. Specifically, we measure the feature similarity between two herbs as:

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{\|\mathbf{h}_i\|_2 \|\mathbf{h}_j\|_2} \quad (9)$$

where $\cos(\dots)$ denotes the cosine similarity, $\mathbf{h}_i$ and $\mathbf{h}_j$ are the learned embeddings of the i th and j th herbs, respectively, and $\|\cdot\|_2$ represents the L_2 norm.

To further incorporate toxic label relationships, we define the label semantic similarity β_{ij} as the fraction of shared toxic labels between two herbs:

$$\beta_{ij} = \frac{\mathbf{y}_i^T \cdot \mathbf{y}_j}{\mathbf{y}_i^T \cdot \mathbf{y}_i} \quad (10)$$

where $\mathbf{y}_i \in \{0, 1\}^L$ and $\mathbf{y}_j \in \{0, 1\}^L$ are the multi-hot toxic label vectors for the i th and j th herbs, respectively, and L is the number of toxic labels. The numerator, $\mathbf{y}_i^T \cdot \mathbf{y}_j$, counts the number of shared toxic labels between the two herbs, while the denominator, $\mathbf{y}_i^T \cdot \mathbf{y}_i$, represents the total number of toxic labels for the i th herb. This ratio-based semantic similarity enables the model to distinguish not only completely identical or disjoint label sets in single-label setting, but also to quantify the partial similarity unique to multi-label scenario.

During subsequent experiments, we found that β may not sufficiently suitable to act as the dynamic coefficient for contrastive learning, as it does not provide a robust measure of relevance between two herbs. In particular, β can assign an excessively large coefficient to herb samples with low toxic label similarity. This issue can degrade the performance of herb toxicity prediction. As a result, we introduce a tunable hyperparameter t to adjust β to obtain a more appropriate dynamic coefficient δ :

$$\delta_{ij} = (\beta_{ij})^t \quad (11)$$

The contrastive learning loss with dynamic coefficient δ_{ij} for a herb pair is defined as:

$$\mathcal{L}_{ij}^{con} = -\delta_{ij} \log \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau)}{\sum_{k=1, k \neq i}^N \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau)} \quad (12)$$

where τ is the temperature hyperparameter used to scale the similarity scores, and N is the total number of herbs in the batch. This formulation allows the model to emphasize similar herb pairs with higher toxicity similarity while effectively learning representations for herbs with multiple related toxic labels. This formulation explicitly leverages the multi-label toxicity of herbs, since each herb pair’s contribution to the loss is scaled by the extent of their shared toxic labels.

For a given pair of herbs $(\mathbf{h}_i, \mathbf{h}_j)$, a higher semantic similarity β_{ij} results in a larger dynamic coefficient δ_{ij} , which in turn leads to a larger loss $\mathcal{L}_{ij}^{con}$. Consequently, the feature similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized toward a higher value. Conversely, if two samples do not share any labels ($\beta_{ij} = \delta_{ij} = 0$), the value $\mathcal{L}_{ij}^{con}$ becomes zero, and their cosine similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized to a lower value. The contrastive learning loss with dynamic coefficient for N herbs is computed as:

$$\mathcal{L}^{con} = \sum_{i=1}^N \sum_{j=1, j \neq i}^N \mathcal{L}_{ij}^{con} \quad (13)$$

3.4. Herb toxicity prediction based on weighted label fusion

For the i th herb, its representation $\mathbf{h}_i$ is passed through a three-layer MLP, followed by a sigmoid activation function, to compute the probability vector for all toxic labels:

$$\tilde{\mathbf{y}}_i^{mlp} = \text{sigmoid}(\text{MLP}(\mathbf{h}_i)) \quad (14)$$

where $\text{MLP}(\cdot)$ denotes a three-layer MLP. $\tilde{\mathbf{y}}_i^{mlp} \in \{0, 1\}^L$ represents the predicted probability vector that the i th herb is associated with toxic label.

However, the above MLP may not fully utilize the valuable knowledge encoded in the training data. To further enhance the toxicity prediction, we introduce a weighted label fusion (WLF) strategy. For each target herb, we retrieve training herbs with similar representations. By virtue of contrastive learning, these neighbors are more likely to share meaningful toxicity patterns with the target herb. HerbToxNet aggregates the toxic labels of neighborhood herbs in a weighted manner, which complements the MLP’s output and contributes to more accurate toxicity prediction.

Let $\mathcal{D} = \{(\mathbf{h}_j, \mathbf{y}_j)\}_{j=1}^N$ denote the training set with N herbs. Given the i th target herb with representation $\mathbf{h}_i$, we retrieve its relevant neighbors from the training set by computing the cosine similarity between $\mathbf{h}_i$ and all $\mathbf{h}_j$ in $\mathcal{D}$. Neighbors whose similarity exceeds a predefined threshold σ are selected to form the neighbor set:

$$\mathcal{N}_\sigma = \{(\mathbf{h}_j, \mathbf{y}_j) \mid \cos(\mathbf{h}_i, \mathbf{h}_j) \geq \sigma\} \quad (15)$$

Instead of using K-nearest neighbors, our threshold-based strategy adaptively selects neighbors that are close to the target herb in the representation space.

The WLF-based prediction for the i th herb is obtained by aggregating the label vectors of the selected neighbors, weighted by their similarities with respect to the target herb:

$$\tilde{\mathbf{y}}_i^{wlf} = \sum_{(\mathbf{h}_j, \mathbf{y}_j) \in \mathcal{N}_\sigma} \mu_j \mathbf{y}_j \quad (16)$$

where the weight μ_j is defined as:

$$\mu_j = \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau')}{\sum_{(\mathbf{h}_k, \mathbf{y}_k) \in \mathcal{N}_\sigma} \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau')} \quad (17)$$

with τ' being a temperature hyperparameter that controls the sharpness of the weight distribution. In this way, neighbors with higher similarity to the i th herb are assigned with larger weights, which more significantly contribute to the WLF-based prediction than other neighbors.

To this end, we combine the MLP-based prediction $\tilde{\mathbf{y}}_i^{mlp}$ with WLF-based prediction $\tilde{\mathbf{y}}_i^{wlf}$ as:

$$\tilde{\mathbf{y}}_i = \alpha \tilde{\mathbf{y}}_i^{wlf} + (1 - \alpha) \tilde{\mathbf{y}}_i^{mlp} \quad (18)$$

where the scalar parameter α balances the two sources of predictions.

We adopt the asymmetric loss (ASL) (Ridnik et al., 2021) as the classification loss to address the imbalanced nature of multi-label herb toxicity prediction. ASL effectively mitigates the impact of label imbalance by assigning different penalties to positive and negative samples, thereby enhancing the model’s focus on rare but critical toxic labels. The loss

函数 被 定义 为 如下:

$$\mathcal{L}^{asl} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L \left[y_{ij} (1 - \tilde{y}_{ij})^{\gamma^+} \log(\tilde{y}_{ij}) + (1 - y_{ij}) (\rho_{ij}^m)^{\gamma^-} \log(1 - p_{ij}^m) \right] \quad (19)$$

$$p_{ij}^m = \max(\tilde{y}_{ij} - m, 0) \quad (20)$$

其中 γ^+ 和 γ^- 分别是 参数 用于 正 和 负 样本的聚焦系数, 而 $m \geq 0$ 是 概率 边界。

HerbToxNet 的 整体 损失 函数 由 ASL 分类 损失 和 对比学习 损失 组成, 公式化表示为:

$$\mathcal{L} = \mathcal{L}^{asl} + \lambda \mathcal{L}^{con} \quad (21)$$

其中 λ 是 一个 超参数 用于 平衡 这两个 损失。算法 1 总结了 HerbToxNet 的 伪 代码

算法1 HerbToxNet整体流程。输入：草药药理学 $\mathbf{p}_i$ 、功效 $\mathbf{e}_i$ ；毒性标签 $\mathbf{y}_i$ ；元路径集合 $\mathcal{P}$ ；节点集 $\mathcal{V}$ ；边集 $\mathcal{E}$ ；模型参数 Θ ；输出：预测的草药毒性标签 $\tilde{\mathbf{y}}$ ；1: $\triangleright$ 异质图构建 2: 构建节点集 $\mathcal{V}_h$ （草药）、 $\mathcal{V}_c$ （成分）、 $\mathcal{V}_t$ （靶点）和边集 $\mathcal{E}_{hc}$ 、 $\mathcal{E}_{ht}$ 、 $\mathcal{E}_{ct}$ 、 $\mathcal{E}_{hh}$ ；3: 对于每个草药, 通过公式(1)初始化节点特征 $\mathbf{f}_i$ ；4: 组装异质图 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ ；5: $\triangleright$ 通过HAN进行草药表示学习 6: 对每个元路径 $p_k \in \mathcal{P}$ 执行 7: 从 $\mathcal{G}$ 中提取元路径子图 $\mathcal{G}_{p_k}$ ；8: 对每个节点 执行 9: 通过公式(2)计算注意力分数 $e_{ij}^{G_{p_k}}$ 并通过公式(3)归一化；10: 通过公式(4)获得邻居聚合特征 $\mathbf{f}_i^{G_{p_k}}$ ；11: 通过公式(5)获得多头特征表示 $\mathbf{z}_i^{G_{p_k}}$ ；12: 结束循环 13: 结束循环 14: 通过公式(6)计算元路径注意力 $w_{G_{p_k}}$ 并通过公式(7)归一化；15: 通过公式(8)融合元路径嵌入得到 $\mathbf{h}_i$ ；16: $\triangleright$ 带动态系数的对比学习 17: 对每个草药对 (i, j) 执行 18: 通过公式(9)计算特征相似度；19: 通过公式(10)和公式(11)计算语义相似度 β_{ij} 和动态系数 δ_{ij} ；20: 结束循环 21: 通过公式(13)计算总对比损失 $\mathcal{L}^{con}$ ；22: $\triangleright$ 草药毒性预测 23: 对每个草药 执行 24: 通过公式(14)用MLP预测 $\tilde{\mathbf{y}}_i^{mlp}$ ；25: 通过公式(18)将 $\tilde{\mathbf{y}}_i^{mlp}$ 和 $\tilde{\mathbf{y}}_i^{wlf}$ 加权融合为 $\tilde{\mathbf{y}}_i$ ；26: 结束循环 27: 通过公式(19)和公式(20)计算非对称损失 $\mathcal{L}^{asl}$ ；28: 通过公式(21)计算总损失 $\mathcal{L}$ ；29: 通过反向传播和Adam优化器更新模型参数 Θ ；30: 返回草药 $\tilde{\mathbf{y}}$ 的预测毒性标签；

4. 实验

4.1. 实验 设置

4.1.1. 数据集

我们 从 已建立的 数据库 中 收集了 草药 的 药理学 特性、 功效 和 相互作用 数据 , 包括 《中国药典》（2020年版）

（Xu 等人, 2021）、HERB（Fang 等人, 2021）、SymMap v2数据库（Wu 等人, 2019b）、TCMBank（Lv 等人, 2023）和 HIT 2.0数据库（Yan 等人, 2022b）。毒性标签主要从谷歌学术、中国知网、维普、万方、PubMed 和 Web of Science 等文献和数据库来源获取, 并分为五类: 心脏毒性、肾毒性、肝毒性、全身毒性和其他毒性。表1给出了数据来源的概览。

我们的数据集构建流程详见补充文件第一节。随后, 我们将已知的草药毒性标签作为该草药的正样本, 其余作为负样本。我们保留15% 的草药作为测试集, 其余草药用于五折交叉验证。该过程独立重复十次以确保稳健性。为保证完整性, 在划分过程中所有与草药相关的关联都被归入同一子集。

4.1.2. 评估 指标

为评估HerbToxNet及基线方法的性能, 我们采用四种广泛使用的评估指标（Qiu等人, 2023）: 宏平均F1分数（Macro-F1）、微平均F1分数（Micro-F1）、平均受试者工作特征曲线下面积（AvgAUC）和单错误率（OneError）。Macro-F1独立计算每个标签的F1分数后取平均值, 反映所有标签的整体性能。Micro-F1聚合所有标签的贡献计算平均F1分数, 反映所有样本的整体性能。AvgAUC计算每个标签的曲线下面积后取平均值, 值越高表示毒性与非毒性标签的可分离性越好。OneError统计预测排名最高的毒性标签不在真实标签中的次数; 值越低表明模型越可能将真实标签排在首位。为与其他指标保持一致, 我们使用1 - OneError代替原值。

4.1.3. 基线 方法

为验证我们提出的HerbToxNet的有效性, 选取了若干代表性方法作为基线, 这些方法可归纳为三类: 草药毒性预测方法、化学药物毒性预测方法以及生物医学领域的多标签预测方法。（一）草药毒性预测方法: ZF-LightGBM（Yu等人, 2025）在正清风痛宁诱导的肝损伤数据集上比较了十种机器学习模型, 发现LightGBM算法表现最佳; QSAR-ANN和QSAR-SVM（Sun等人, 2019）利用草药肾毒性数据构建定量构效关系模型, 并分别应用人工神经网络和支持向量机算法预测肾毒性; EI-MS-RF（Jia等人, 2024）通过将草药毒性与电子电离质谱数据的高级分析描述符相关联, 结合可解释的机器学习模型进行毒性预测。（二）化学药物毒性预测方法: DeepDILI（Li等人, 2020）将传统机器学习算法生成的模型级表示与基于Mold2描述符的深度学习框架相结合, 以预测药物性肝损伤; R-E-GA（Yan等人, 2022a）在特征空间中对高维分子表示的融合向量进行旋转以寻找具有更好预测性能的特征子集, 随后将基于Adaboost算法的集成学习方法整合到R-E-GA中以提高预测精度; TIRPnet（Wei等人, 2024）旨在从海量的中医药自发报告中解读知识, 实现对单一成分潜在在不良反应（ADRs）的预测。（三）多标签预测方法: ML-PRDF（Gong等人, 2023）首先利用PCC-MLRF算法（Gong等人, 2023）进行特征选择, 然后将精心筛选的特征输入多标签深度森林模型以进行区分和预测分析; MulGrAL（Pham等人, 2022）构建与预测结果相关的异质信息网络, 随后组合多个多标签分类器以提升预测的准确性和可靠性; MLC-CNN（Ge等人, 2021）通过根据标签间相关性对多标签进行分组, 并利用神经网络模型对

function is defined as follows:

$$\mathcal{L}^{asl} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L \left[y_{ij} (1 - \tilde{y}_{ij})^{\gamma^+} \log(\tilde{y}_{ij}) + (1 - y_{ij}) (\rho_{ij}^m)^{\gamma^-} \log(1 - p_{ij}^m) \right] \quad (19)$$

$$p_{ij}^m = \max(\tilde{y}_{ij} - m, 0) \quad (20)$$

where γ^+ and γ^- are focusing parameters for positive and negative samples, respectively, and $m \geq 0$ is the probability margin.

The overall loss function of HerbToxNet consists of the ASL classification loss and the contrastive learning loss, formulated as:

$$\mathcal{L} = \mathcal{L}^{asl} + \lambda \mathcal{L}^{con} \quad (21)$$

where λ is a the hyperparameter that balances the two losses. Algorithm 1 summarizes the pseudo code of HerbToxNet

Algorithm 1 The overall process of HerbToxNet.

Input: herb pharmacology $\mathbf{p}_i$, efficacy $\mathbf{e}_i$; toxic labels $\mathbf{y}_i$; meta-path set $\mathcal{P}$; node sets $\mathcal{V}$; edge sets $\mathcal{E}$; model parameters Θ ;

Output: Predicted herb toxic label $\tilde{\mathbf{y}}$;

```

1:  $\triangleright$  Heterogeneous Graph Construction
2:   Build node set  $\mathcal{V}_h$  (herb),  $\mathcal{V}_c$  (ingredient),  $\mathcal{V}_t$  (target) and edge set  $\mathcal{E}_{hc}, \mathcal{E}_{ht}, \mathcal{E}_{ct}, \mathcal{E}_{hh}$ ;
3:   For each herb  $i$ , initialize node features  $\mathbf{f}_i$  via Eq. (1);
4:   Assemble the heterogeneous graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ ;
5:  $\triangleright$  Herb Representation Learning by HAN
6: for each meta-path  $p_k \in \mathcal{P}$  do
7:   Extract meta-path subgraph  $\mathcal{G}_{p_k}$  from  $\mathcal{G}$ ;
8:   for each node  $i$  do
9:     Compute attention score  $e_{ij}^{G_{p_k}}$  via Eq. (2) and normalize it via Eq. (3);
10:    Get neighbor-aggregated features  $\mathbf{f}_i^{G_{p_k}}$  via Eq. (4);
11:    Obtain multi-head feature representation  $\mathbf{z}_i^{G_{p_k}}$  via Eq. (5);
12:  end for
13: end for
14:   Compute meta-path attention  $w_{G_{p_k}}$  via Eq. (6) and normalize it via Eq. (7);
15:   Fuse meta-path embeddings to get  $\mathbf{h}_i$  via Eq. (8);
16:  $\triangleright$  Contrastive Learning with Dynamic Coefficient
17: for each herb pair  $(i, j)$  do
18:   Compute feature similarity via Eq. (9);
19:   Compute semantic similarity  $\beta_{ij}$  and dynamic coefficient  $\delta_{ij}$  via Eq. (10) and Eq. (11);
20: end for
21:   Compute total contrastive loss  $\mathcal{L}^{con}$  via Eq. (13);
22:  $\triangleright$  Herb Toxicity Prediction
23: for each herb  $i$  do
24:   MLP predicts  $\tilde{\mathbf{y}}_i^{mlp}$  via Eq. (14);
25:   Weighted fuse  $\tilde{\mathbf{y}}_i^{mlp}$  and  $\tilde{\mathbf{y}}_i^{wlf}$  into  $\tilde{\mathbf{y}}_i$  via Eq. (18);
26: end for
27:   Compute asymmetric loss  $\mathcal{L}^{asl}$  via Eq. (19) and Eq. (20);
28:   Compute total loss  $\mathcal{L}$  via Eq. (21);
29:   Update model parameters  $\Theta$  by back propagation and Adam;
30: return Predicted toxic labels of herb  $\tilde{\mathbf{y}}$ ;
```

4. Experiment

4.1. Experimental setup

4.1.1. Dataset

We collected pharmacological properties, efficacy, and interaction data for herbs from established databases, including ChP (2020)

(Xu et al., 2021), HERB (Fang et al., 2021), SymMap v2 (Wu et al., 2019b), TCMBank (Lv et al., 2023), and HIT 2.0 (Yan et al., 2022b). Toxicity labels were obtained from major literature and database sources such as Google Scholar, CNKI, VIP, WanFang, PubMed, and Web of Science, and were grouped into five categories: cardiotoxicity, nephrotoxicity, hepatotoxicity, systemic toxicity, and other toxicity. Table 1 gives an overview of the data sources.

The curation pipeline of our dataset is reported in Section I of Supplementary file. Then, we take known toxic labels of a herb as positive samples and others as negative ones of this herb. We keep 15% of the herbs as the testing set, while the remaining herbs are used for five-fold cross-validation. This process is independently repeated ten times to ensure robustness. To ensure the integrity, all interactions associated with a herb are grouped into the same subset during the partition process.

4.1.2. Evaluation metrics

To evaluate the performance of HerbToxNet and the baseline methods, we employed four widely used evaluation metrics (Qiu et al., 2023): Macro-averaged F1 Score (Macro-F1), Micro-averaged F1 Score (Micro-F1), Average Area Under the receiver operating characteristic Curve (AvgAUC), and One Error (OneError). Macro-F1 independently calculates the F1 score for each label and then takes the average, reflecting the overall performance across all labels. Micro-F1 aggregates the contributions of all labels to compute the average F1 score, reflecting overall performance on all samples. AvgAUC computes the AUC for each label and then takes the average, a higher value means a better separability between toxic and non-toxic labels. OneError counts the number of the predicted toxic label with top-rank is not among the true labels; a lower values indicate that the model is more likely to rank a true label first. For consistency with other metrics, we use 1 - OneError instead.

4.1.3. Baseline methods

To prove the effectiveness of our proposed HerbToxNet, several representative methods were selected as the baseline, which can be summarized into three categories: herb toxicity prediction methods, chemical drug toxicity prediction methods, and multi-label prediction methods in the field of biomedicine. (i) Herb toxicity prediction methods: ZF-LightGBM (Yu et al., 2025) compares ten machine learning models on *Zhengqing Fengtongning*-induced liver injury dataset and finds LightGBM performing the best; QSAR-ANN and QSAR-SVM (Sun et al., 2019) uses herb nephrotoxicity data to construct QSAR model, and apply ANN and SVM algorithms to predict nephrotoxicity, respectively; EI-MS-RF (Jia et al., 2024) makes toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. (ii) Chemical drug toxicity prediction methods: DeepDILI (Li et al., 2020) combines model-level representations generated by traditional machine learning algorithms with a deep learning framework based on Mold2 descriptors to predict drug-induced liver injury; R-E-GA (Yan et al., 2022a) rotates the fusion vector represented by a high-dimensional molecule in the feature space to find a subset of features with better predictive performance, and then integrates an Adaboost-based ensemble learning method into R-E-GA to improve prediction accuracy; TIRPnet (Wei et al., 2024) is designed to decipher knowledge from abundant of TCM spontaneous reports, enabling the prediction of potential adverse drug reactions (ADRs) for individual ingredient. (iii) multi-label prediction methods: ML-PRDF (Gong et al., 2023) initially utilizes PCC-MLRF (Gong et al., 2023) for feature selection, and then inputs the meticulously selected features into the multi-label deep forest model to conduct differentiation and predictive analysis; MulGrAL (Pham et al., 2022) constructs a heterogeneous information network relevant to the prediction outcomes, subsequently combines multiple multi-label classifiers to enhance the accuracy and reliability of prediction; MLC-CNN (Ge et al., 2021) involves grouping multi-labels according to their inter-label correlations and utilizes neural network models to perform predictions for

表 1

Details of the collected data sources.			
Type	Size	来源	
Herb	Pharmacological Property	252×33	Chp (2020), HERB, SymMap v2, TCM- 银行, HIT 2.0
	功效	252×331	
	草药-靶点	34988	
	Herb-成分成分-靶点	3275	
毒性		5932	
	心脏毒性	74	Google Scholar CNKI VIP WanFang , , , , PubMed WOS
	肾毒性 肝毒性	101	
		114	
	系统 ic Toxicity	167	
	其他 Toxicity	150	

注: The size of ‘Efficacy’ is 252×331 indicating that the number of herbs is 252 and the number of efficacy is 331. ‘Pharmacological Property’ is represented in the same way as ‘功效’。

每个 组别 进行 预测 , 有效 利用 了 多样化 特征 对 不同<style id='23'> 标签<style id='25'> 的影响。

由于前两类方法最初是为二分类任务设计的, 我们修改了其最终的二值输出以生成适用于本任务的多标签预测。此外, 由于我们的数据集中没有EI-MS数据, 我们使用草药药理特性和功效数据替代EI-MS描述符, 并按照原始研究所述实施了相同的EI-MS-RF流程。对于每个基线方法, 我们以推荐参数为起点, 并根据我们的数据集进行了必要的微调。所有基线方法和我们的HerbToxNet的详细参数, 以及HerbToxNet的全面参数敏感性分析, 均在补充文件第III节中呈现。

4.2. 与 现有 解决方案的 结果 比较

表 2 总结了 HerbToxNet 与 基线方法的性能表现。HerbToxNet 在大多数 评估指标中 实现了 最佳的 草药 毒性 预测 性能。其他 重要 观察 结果 包括: (i) HerbToxNet在草药毒性预测方面显著优于其他方法, 包括 ZF-LightGBM、QSAR-SVM、EI-MS-RF和QSAR-ANN, 这主要源于后者无法有效捕捉草药复杂的多成分和多靶点特征。这一缺陷限制了它们的泛化能力, 在处理多标签毒性预测任务时尤为明显。此外, 这些方法通常仅预测单一毒性标签, 且未能有效建模毒性标签间的复杂关联, 从而限制了它们在多标签毒性预测中的表现。相比之下, HerbToxNet在复杂多标签毒性预测任务中取得了更优性能, 在多数评估指标上显著领先。其优势体现在: 首先, HerbToxNet构建了草药-成分-靶点异质网络, 并利用元路径提取草药属性与关

联信息, 实现了对草药复杂特征的精准建模。其次, 它引入了带动态系数的对比学习策略, 使具有相似毒性标签的草药表示相互靠近, 而不同毒性标签的表示相互远离, 增强了模型对多标签任务的适应性。第三, HerbToxNet采用加权标签融合策略, 有效利用相似草药的毒性标签, 从而进一步提升预测性能并缓解草药数据稀缺的影响。(ii) HerbToxNet同样显著优于药物毒性预测方法 (包括DeepDILI、R-E-GA和TIRPnet)。这些方法在草药多标签毒性预测任务中表现不佳, 因为化学药物通常具有明确的分子结构和指纹, 可为预测提供丰富详细信息。而草药由多成分构成, 其药理学数据维度较低, 使用独热编码表示这些成分不足以捕捉其复杂特征。此外, 这些方法可能无法有效捕捉和利用草药相关毒性标签间的关联。HerbToxNet通过异质网络建模、带动态系数的对比学习以及加权标签融合策略应对这些挑战。这些创新使 HerbToxNet能有效捕捉草药的复杂特征, 并显著提升其在多标签毒性预测任务中的性能。

(iii) 与其他两类方法相比, 多标签分类方法在预测草药毒性标签方面展现出一定优势。这是因为它们能够同时考虑多个毒性标签, 天然适用于多标签草药毒性预测。尽管这些方法在某些场景下能提供有竞争力的结果, 但在挖掘有限草药数据的内在信息和复杂特征方面可能受限。因此, 如表 2所示, 这些方法再次输给了HerbToxNet。HerbToxNet的卓越性能主要归功于其有效的建模

表 2

Performance comparison of HerbToxNet and baseline methods in predicting herb toxicity.					
	Macro F1	微观 F1	AvgAUC	1 - 单错误率	
ZF-LightGBM	0.237 ±0.04 13	0.4127±0.04 20	0.5163±0.0307	0.2535±0.0487	
QSAR-ANN	0.4195±0.0460	0.4510±0.0429	0.5105±0.0343	0.1936±0.0536	
ML-PRDF模型	0.1619±0.0235	0.3842±0.04 22	0.5132±0.04 22	0.2582±0.0520	
EI-MS-RF模型	0.3432±0.0384	0.4437±0.0304	0.5120±0.0370	0.2242±0.048 2	
DeepDILI	0.3035±0.0956	0.4756±0.0842	0.5166±0.0489	0.5329±0.070 2	
R-E-GA	0.2894±0.0470	0.4220±0.0465	0.5092±0.0352	0.2313±0.0417	
TIRPnet模型	0.4089±0.0420	0.4265±0.0468	0.5058±0.0384	0.2111±0.0608	
ML-PRDF模型	0.4059±0.0327	0.4508±0.0439	0.5294±0.0548	0.4711±0.0698	
MulGRaL	0.4135±0.0421	0.5576±0.0325	0.5606±0.0209	0.5195±0.0611	
MLC-CNN模型	0.4966±0.0653	0.5086±0.0508	0.5525±0.0349	0.5115±0.0676	
	0.5652± 0.0161	0.6247±0.0172	0.6003± 0.0141	0.5587±0.0485	

注: The best results are highlighted in bold font with statistical significance checked , % by student *t*-test at 95 level.

Table 1
Details of the collected data sources.

	Type	Size	Sources
Herb	Pharmacological Property	252×33	Chp (2020), HERB, SymMap v2, TCM-Bank, HIT 2.0
	Efficacy	252×331	
	Herb-Target	34988	
	Herb-Ingredient	3275	
Toxicity	Ingredient-Target	5932	Google Scholar, CNKI, VIP, WanFang, PubMed, WOS
	Cardiotoxicity	74	
	Nephrotoxicity	101	
	Hepatotoxicity	114	
	Systemic Toxicity	167	
	Other Toxicity	150	

Note: The size of ‘Efficacy’ is 252×331, indicating that the number of herbs is 252 and the number of efficacy is 331. ‘Pharmacological Property’ is represented in the same way as ‘Efficacy’.

each group, effectively leveraging the impact of diverse features on distinct labels.

Since the first two categories of methods were originally developed for binary classification tasks, we modify their final binary outputs to produce multi-label predictions suitable for our task. In addition, as EI-MS data are not available in our dataset, we substitute the EI-MS descriptors with herb pharmacological property and efficacy data, and implement the same EI-MS-RF pipeline as described in the original work. For each baseline method, we used the recommended parameters as a starting point and fine-tuned them as necessary for our dataset. Detailed parameters for all baselines and our HerbToxNet, as well as a comprehensive parameter sensitivity analysis of HerbToxNet, are presented in Section III of Supplementary file.

4.2. Results comparison with existing solutions

Table 2 summarizes the performance of HerbToxNet and baselines. HerbToxNet achieves the best herb toxicity prediction performance in most evaluation metrics. Other important observations are: (i) HerbToxNet significantly outperforms other herb toxicity prediction methods, including ZF-LightGBM, QSAR-SVM, EI-MS-RF, and QSAR-ANN, primarily due to they inability to effectively capture the complex multi-ingredient and multi-target characteristics of herbs. This defect limits their generalization ability, which is particularly evident when handling multi-label toxicity prediction tasks. Moreover, they typically predict single toxic labels and fail to effectively model the intricate correlations between toxic labels, thereby restricting their performance in multi-label toxicity prediction. In contrast, HerbToxNet achieves better performance in complex multi-label toxicity prediction tasks, significantly leading in most evaluation metrics. Its advantages are as follows. First, HerbToxNet constructs a herb-ingredient-target heterogeneous network and utilizes meta-path to extract properties and rela-

Table 2
Performance comparison of HerbToxNet and baseline methods in predicting herb toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.2371±0.0413	0.4127±0.0420	0.5163±0.0307	0.2535±0.0487
QSAR-ANN	0.4195±0.0460	0.4510±0.0429	0.5105±0.0343	0.1936±0.0536
QSAR-SVM	0.1619±0.0235	0.3842±0.0422	0.5132±0.0422	0.2582±0.0520
EI-MS-RF	0.3432±0.0384	0.4437±0.0304	0.5120±0.0370	0.2242±0.0482
DeepDILI	0.3035±0.0956	0.4756±0.0842	0.5166±0.0489	0.5329±0.0702
R-E-GA	0.2894±0.0470	0.4220±0.0465	0.5092±0.0352	0.2313±0.0417
TIRPnet	0.4089±0.0420	0.4265±0.0468	0.5058±0.0384	0.2111±0.0608
ML-PRDF	0.4059±0.0327	0.4508±0.0439	0.5294±0.0548	0.4711±0.0698
MulGRaL	0.4135±0.0421	0.5576±0.0325	0.5606±0.0209	0.5195±0.0611
MLC-CNN	0.4966±0.0653	0.5086±0.0508	0.5525±0.0349	0.5115±0.0676
	0.5652± 0.0161	0.6247±0.0172	0.6003± 0.0141	0.5587±0.0485

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

表 3

Comparison of HerbToxNet with other methods on hepatotoxicity and nephrotoxicity prediction.

	H肝毒性	Pr预测		Nephro毒性	Pr预测	
		AUROC	AUPRC		AUROC	AUPRC
ZF-Li ghtGBM	0.5610±0.0803	0.5369±0.0944	0.6323±0.0565	0.5682±0.0554	0.4668±0.0723	0.5881±0.0577
QSAR-ANN	0.5543±0.0825	0.5409±0.1038	0.6427±0.0565	0.5571±0.0738	0.4847±0.0798	0.5973±0.0609
M 宏量网络关系, 支持网络	0.5675±0.1038	0.5753±0.1197	0.6295±0.0574	0.5695±0.0649	0.4913±0.0695	0.5848±0.0585
	0.5624±0.0765	0.5304±0.0984	0.6435±0.0594	0.5431±0.0743	0.4529±0.0782	0.6023±0.0566
EI-MS-RF模型	0.5486±0.0763	0.5658±0.1005	0.6272±0.0580	0.5799±0.0744	0.4989±0.0782	0.5949±0.0550
DeepDILI	0.5342±0.0786	0.5437±0.1011	0.6218±0.0574	0.5627±0.0804	0.5053±0.0939	0.5728±0.0578
R-E-GA	0.5269±0.0824	0.4959±0.0907	0.5084±0.0918	0.4949±0.0722	0.4279±0.0735	0.4630±0.0761
TIRPnet模型	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293
HerbToxNet	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293

注： The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

对 草药、成分和靶点通过异质网络、对比学习和加权标签融合进行的建模。

4.3. 肝毒性与肾毒性预测

HerbToxNet专为多标签毒性预测而设计，而大多数方法针对草药（Jia 等人，2024；Sun 等人，2019；Yu 等人，2025）和药物（Li 等人，2020；Wei 等人，2024；Yan 等人，2022a）的单标签毒性预测，它们基于单一类型毒性的二元任务。为评估HerbToxNet能否适配这些二元任务，我们从数据集中分离出肝毒性和肾毒性草药，将其划分为肝毒性和肾毒性数据集，并在两个数据集中评估我们的方法。其中，肝毒性数据集包含114种草药，肾毒性数据集包含101种草药。由于这是二元任务，采用AUROC、AUPRC和F1分数进行评估。

如表3所示，尽管HerbToxNet最初是为多标签毒性预测设计的，但在应用于针对个体毒性类型的二分类任务时，它展现出了良好的适应性和鲁棒性。在肝毒性预测任务中，HerbToxNet在所有评估指标上均取得最佳性能，优于所有基线模型。与现有方法（如Deep-DILI、基于QSAR的方法和EI-MS-RF模型）相比，HerbToxNet获得了更高的AUPRC值，表明其在类别不平衡设置下识别阳性样本的能力更强。同样，在肾毒性预测任务中，HerbToxNet保持了竞争优势，取得了最高的AUROC、AUPRC和F1分数，这证明了其在识别肾毒性草药方面强大的判别能力。值得注意的是，HerbToxNet在两种毒性类型上均获得了一致更好的结果，验证了其在不同毒理学终点间稳定的适应性和学习能力。

这些结果凸显了HerbToxNet不仅能够有效地建模复杂的多标签毒性关系，而且还能提取关键特征并评估二元分类任务中的毒性风险。

4.4. 药物毒性预测

HerbToxNet不仅限于草药毒性预测，还可扩展应用于化学药物。为研究其泛化能力，我们在UniTox数据集（Silberg等人，2024）上进行了评估，该数据集包含1349种标注了八种毒性标签的化学化合物。考虑到化学药物与草药在结构和语义上的差异，我们移除了异质图和HAN，直接使用每种药物的SMILES表示作为输入。从SMILES字符串中，我们提取摩根指纹来表示分子特征。这些特征随后与对比学习模块和加权标签融合策略相结合，以执行化学药物的毒性预测。在基线方法中，MulFRaL被排除在比较之外，因为它严重依赖异质图结构，而该结构在此场景中不适用。所有其他基线方法保持不变。实验设置和评估指标与草药毒性预测任务中使用的相同。

表4总结了HerbToxNet和基线方法在预测药物毒性方面的性能表现。在大多数评估指标中，HerbToxNet的结果均优于基线方法。它在化学药物的毒性预测任务中再次展现出卓越性能，这主要归功于其在特征表示和标签预测两方面的方法论优势。尽管此设置中排除了异质图和HAN，但HerbToxNet利用对比学习模块增强了药物表示的判别能力。具体而言，对比学习策略促使具有相似毒性特征的药物表示在特征空间中更加接近，这有助于区分不同的毒性标签。Morgan指纹的使用提供了一种紧凑且广泛适用的分子表示，即使在没有中医药特定复杂语义上下文的情况下，该模型仍能保持较高的预测效能。此外，WLF融合了来自多层感知器的预测结果和从相似草药聚合的预测结果，有效缓解了训练数据有限的问题。这些结果不仅验证了HerbToxNet的泛化能力，而且

表 4

Performance comparison of HerbToxNet and baseline methods in predicting drug toxicity.

	Macro F1	微观 F1	AvgAUC	1 - 单错误率
ZF-Li ghtGBM	0.6107±0.0169	0.6961±0.0114	0.6973±0.0139	0.3610±0.0226
QSAR-ANN	0.6196±0.0155	0.7160±0.0128	0.7059±0.0163	0.3250±0.0223
M 宏量网络关系, 支持网络	0.5241±0.0212	0.7085±0.0109	0.7038±0.0172	0.3424±0.0281
	0.5696±0.0155	0.7129±0.0110	0.7038±0.0148	0.3689±0.0245
EI-MS-RF模型	0.5929±0.0174	0.7184±0.0132	0.7055±0.0173	0.8638±0.0234
DeepDILI	0.5382±0.0360	0.6538±0.0303	0.5998±0.0055	0.3714±0.0186
R-E-GA	0.5861±0.0383	0.6905±0.0162	0.6618±0.0226	0.3658±0.0563
TIRPnet	0.6107±0.0159	0.7080±0.0117	0.6260±0.0129	0.8504±0.0149
ML-PRDF	0.5091±0.0682	0.6648±0.0301	0.6179±0.0265	0.8718±0.0186
MLC-CNN	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164
HerbToxNet	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164

注： The best results are highlighted in bold font with statistical significance checked , % by student *t*-test at 95 % level.

Table 3

Comparison of HerbToxNet with other methods on hepatotoxicity and nephrotoxicity prediction.

	Hepatotoxicity Prediction			Nephrotoxicity Prediction		
	AUROC	AUPRC	F1	AUROC	AUPRC	F1
ZF-LightGBM	0.5610±0.0803	0.5369±0.0944	0.6323±0.0565	0.5682±0.0554	0.4668±0.0723	0.5881±0.0577
QSAR-ANN	0.5543±0.0825	0.5409±0.1038	0.6427±0.0565	0.5571±0.0738	0.4847±0.0798	0.5973±0.0609
QSAR-SVM	0.5675±0.1038	0.5753±0.1197	0.6295±0.0574	0.5695±0.0649	0.4913±0.0695	0.5848±0.0585
EI-MS-RF	0.5624±0.0765	0.5304±0.0984	0.6435±0.0594	0.5431±0.0743	0.4520±0.0782	0.6023±0.0566
DeepDILI	0.5486±0.0763	0.5658±0.1005	0.6272±0.0580	0.5799±0.0744	0.4989±0.0782	0.5949±0.0550
R-E-GA	0.5342±0.0786	0.5437±0.1011	0.6218±0.0574	0.5627±0.0804	0.5053±0.0939	0.5728±0.0578
TIRPnet	0.5269±0.0824	0.4959±0.0907	0.5084±0.0918	0.4949±0.0722	0.4279±0.0735	0.4630±0.0761
HerbToxNet	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

of herbs, ingredients and targets by heterogeneous network, contrastive learning and weighted label fusion.

4.3. Hepatotoxicity and nephrotoxicity prediction

HerbToxNet is designed for multi-label toxicity prediction, while most methods target at single-label toxicity prediction of herbs (Jia et al., 2024; Sun et al., 2019; Yu et al., 2025) and of drugs (Li et al., 2020; Wei et al., 2024; Yan et al., 2022a), they build on binary tasks for single type toxicity. To assess whether HerbToxNet can be adapted to these binary tasks, we isolated hepatotoxicity and nephrotoxicity herbs from the dataset, divided them as hepatotoxicity and nephrotoxicity datasets and evaluated our approach in both datasets. Among them, the hepatotoxicity dataset contains 114 herbs and the nephrotoxicity dataset contains 101 herbs. Since this is a binary task, AUROC, AUPRC, and F1 are used for evaluation.

As shown in Table 3, although HerbToxNet is originally designed for multi-label toxicity prediction, it demonstrates good adaptability and robustness when applied to binary classification tasks focused on individual toxicity types. In the hepatotoxicity prediction task, HerbToxNet achieves the best performance in all evaluation metrics, outperforming all baseline models. Compared with existing methods such as DeepDILI, QSAR-based approaches, and EI-MS-RF, HerbToxNet achieves higher AUPRC values, suggesting a better capability to identify positive samples under class-imbalanced settings. Similarly, HerbToxNet holds competitive performance in the nephrotoxicity prediction task, achieving the highest AUROC, AUPRC, and F1 scores, which demonstrates its strong discriminative ability in identifying nephrotoxic herbs. Notably, HerbToxNet obtains consistently better results across both toxicity types, validating its stable adaptability and learning capacity across different toxicological endpoints.

These results underscore that HerbToxNet is not only effective in modeling complex multi-label toxicity relationships, but also capable of extracting critical features and assessing toxicity risk in binary classification tasks.

4.4. Drug toxicity prediction

HerbToxNet is not limited to herb toxicity prediction, it can also be extended to chemical drugs. To study its generalizability, we evaluate it on the UniTox dataset (Silberg et al., 2024), which contains 1349 chemical compounds annotated with eight toxic labels. Given the structural and semantic differences between chemical drugs and herbs, we remove the heterogeneous graph and HAN, and directly use the SMILES representation of each drug as input. From the SMILES strings, we extract Morgan fingerprints to represent molecular features. These features are then integrated with the contrastive learning module and the weighted label fusion strategy to perform toxicity prediction for chemical drugs. Among the baseline methods, MulFRaL was excluded from comparison as it relies heavily on heterogeneous graph structures, which are not applicable in this setting. All other baseline methods keep unchanged. The experimental setup and evaluation metrics are the same as those used in the herb toxicity prediction task.

Table 4 summarizes the performance of HerbToxNet and baselines on predicting drug toxicity. The results of HerbToxNet are better than those of baseline methods in most evaluation metrics. It again achieves superior performance in the toxicity prediction task for chemical drugs, primarily due to its methodological strengths in both feature representation and label prediction. Although the heterogeneous graph and HAN are excluded in this setting, HerbToxNet leverages the contrastive learning module to enhance the discriminability of drug representations. Specifically, the contrastive learning strategy encourages representations of drugs with similar toxicity profiles to be closer in the feature space, this helps to distinguish different toxic labels. The use of Morgan fingerprints offers a compact and widely applicable molecular representation, which enables the model to maintain high prediction efficacy even in the absence of complex semantic contexts specific to TCM. Moreover, WLF combines the predictions from MLP and those aggregated from similar herbs, effectively alleviating the issue of limited training data. These results not only validate the generalizability of HerbToxNet, but

Table 4

Performance comparison of HerbToxNet and baseline methods in predicting drug toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.6107±0.0169	0.6961±0.0114	0.6973±0.0139	0.3610±0.0226
QSAR-ANN	0.6196±0.0155	0.7160±0.0128	0.7059±0.0163	0.3250±0.0223
QSAR-SVM	0.5241±0.0212	0.7085±0.0109	0.7038±0.0172	0.3424±0.0281
EI-MS-RF	0.5696±0.0155	0.7129±0.0110	0.7038±0.0148	0.3689±0.0245
DeepDILI	0.5929±0.0174	0.7184±0.0132	0.7055±0.0173	0.8638±0.0234
R-E-GA	0.5382±0.0360	0.6538±0.0303	0.5998±0.0055	0.3714±0.0186
TIRPnet	0.5861±0.0383	0.6905±0.0162	0.6618±0.0226	0.3658±0.0563
ML-PRDF	0.6107±0.0159	0.7080±0.0117	0.6260±0.0129	0.8504±0.0149
MLC-CNN	0.5091±0.0682	0.6648±0.0301	0.6179±0.0265	0.8718±0.0186
HerbToxNet	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

也凸显了其核心设计在不同毒性预测场景中的方法论适应性。

4.5. 消融研究与参数敏感性分析

为深入探究HerbToxNet的贡献因素，我们引入了五种变体：(i) w/o HAN直接使用草药的属性和功效数据作为特征，不构建异质图和元路径，但结合对比学习和加权标签融合策略进行预测。(ii) w/o PE在异质图中随机初始化草药节点特征，而非使用药理学属性和功效数据作为初始特征。(iii) w/o CL在训练阶段移除了带动态系数的对比学习模块。(iv) w/o δ 将动态系数 δ_{ij} 替换为毒性标签相似度 β_{ij} 。(v) w/o WLF仅使用多层感知器进行预测，未结合加权标签融合策略。详细信息记录于补充文件第二节。总体而言，以明显优势超越其变体，凸显了HerbToxNet中每个被移除因素在草药毒性预测中的关键作用。

我们还进行了不同实验来研究HerbToxNet的参数敏感性。结果已在补充文件的第三节中报告并分析。

4.6. 草药毒性预测

在本案例研究中，我们利用HerbToxNet对典型草药（包括柴胡、何首乌、巴豆、附子和蓖麻子）在实际场景中的毒性进行预测分析。为严格评估模型的泛化能力，数据集中所有其他草药均作为训练集，而选定的五种草药作为独立测试集进行毒性预测。将预测毒性与已知毒性标签进行对比，并从PubMed检索相关文献，进一步验证模型预测结果与当前研究结论及传统中医药理论的一致性。

结果表明，HerbToxNet能够准确识别典型肝毒性草药（如柴胡和何首乌）的关键毒性，有效揭示极寒或极热草药（如巴豆和附子）的毒性风险，并发现传统归类为平性草药（如蓖麻子）的潜在毒性。此外，HerbToxNet生成的毒性预测与既有的毒理学证据及经典中医药分类体系基本一致，表明该模型可利用传统草药属性（性、味、归经）中编码的信息来评估毒性风险。

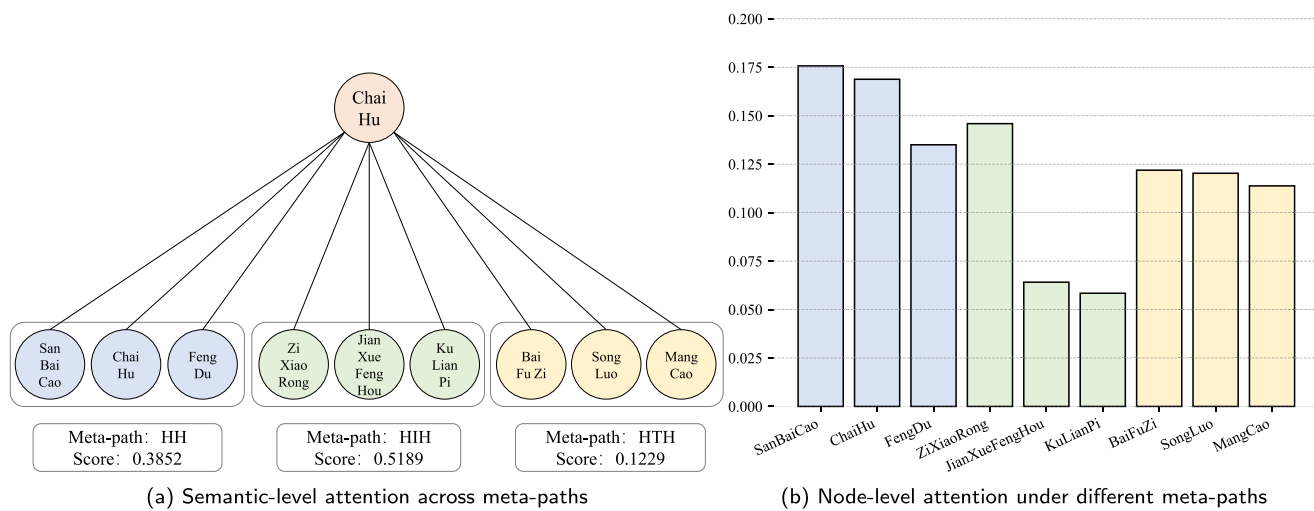
实验流程的详细说明、案例预测结果及全面分析均在补充材料第四部分中提供。

4.7. 可解释性分析

4.7.1. 模型可解释性分析

为捕捉草药、成分与靶点间的复杂关系，我们采用异质注意力网络（HAN）（Wang等人，2019年），该网络因其分层注意力机制而尤为突出。该机制使模型在学习节点嵌入时能同时捕获节点邻近样本的重要性及不同元路径的相关性。为深入理解模型可解释性并更好把握邻近样本与元路径的相对重要性，我们对分层注意力机制进行了详细分析。具体而言，我们以柴胡为例，可视化该节点在不同元路径下与其邻近样本的注意力分数，如图2所示。从语义级分层注意力视角来看，HAN为三条元路径（HH、HH和HTH）分配了不同的注意力权重，反映了模型在表征草药节点柴胡时对不同语义关系的关注程度。其中HH元路径获得最高的语义注意力分数，表明具有共享成分的草药在柴胡特征学习中起主导作用；HH元路径权重适中，而HTH元路径得分最低。这种注意力分数分布与数据集的连通模式相符：HH元路径关联数量较少但更具特异性，使模型能捕获独特的成分级特征；HTH元路径涉及更多连接，导致交互更密集但冗余度更高，削弱了其精确表征草药的能力；HH的中间权重则反映这些草药-草药关联是通过药理学和功效相似性间接推断得出，其特异性低于HH但冗余度低于HTH。

在节点级注意力机制中，我们选择每个元路径下注意力分数最高的前三个节点。对于HH元路径，柴胡与三白草、自身以及丰独相连。这三种草药均具有辛味特性。在归经方面，柴胡和丰独均归肝经。虽然三白草主要归肺经，但肺与肝在生理上相互关联，可实现互补的治疗效果。在功能上，三者均表现出抗炎、消肿和镇痛特性。此外，与自节点的连接



also highlight the methodological adaptability of its core design across different toxicity prediction scenarios.

4.5. Ablation study and parameter sensitivity analysis

To further study the contribution factors of HerbToxNet, we introduce five variants: (i) w/o HAN directly uses the properties and efficacy data of herb as features without constructing heterogeneous graph and the meta-path, but combined with contrastive learning and weighted label fusion strategies for prediction. (ii) w/o PE randomly initializes the herb node features in the heterogeneous graph instead of using pharmacological property and efficacy data as initial features. (iii) w/o CL removes the contrastive learning module with dynamic coefficients in the training phase. (iv) w/o δ replaces the dynamic coefficient δ_{ij} with the toxicity label similarity β_{ij} . (v) w/o WLF only uses the MLP for prediction, without incorporating the weighted label fusion strategy. The details are reported in Section II of Supplementary file. Overall, outperforms its variants by a clear margin, underscoring the vital role of each removed factor in HerbToxNet for the prediction of herb toxicity.

We also conduct different experiments to study the parameter sensitivity of HerbToxNet. The results are reported and analyzed in Section III of Supplementary file.

4.6. Toxicity prediction of herbs

In this case study, we utilized HerbToxNet to predict and analyze the toxicity of canonical herbs, including *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, and *Bi Ma Zi*, in real-world scenarios. To rigorously evaluate the model's generalization capability, all other herbs in the dataset were used as the training set, while the five selected herbs were treated as a separate test set for toxicity prediction. The predicted toxicities were compared with known toxic labels, and relevant literature was retrieved from PubMed to further verify the consistency of model predictions with current research findings and traditional TCM theory.

The results demonstrate that HerbToxNet can accurately identify key toxicities of canonical hepatotoxic herbs (such as *Chai Hu* and *He Shou Wu*), effectively reveal toxicity risks in extremely “cold” or “hot” herbs (such as *Ba Dou* and *Fu Zi*), and uncover potential toxicity in herbs traditionally classified as neutral (“ping”) (such as *Bi Ma Zi*). Moreover, the toxicity predictions generated by HerbToxNet are broadly consistent with both established toxicological evidence and the classical TCM classification system, indicating that the model can leverage information encoded in traditional herb properties (nature, flavor, meridian tropism) to assess toxicity risk.

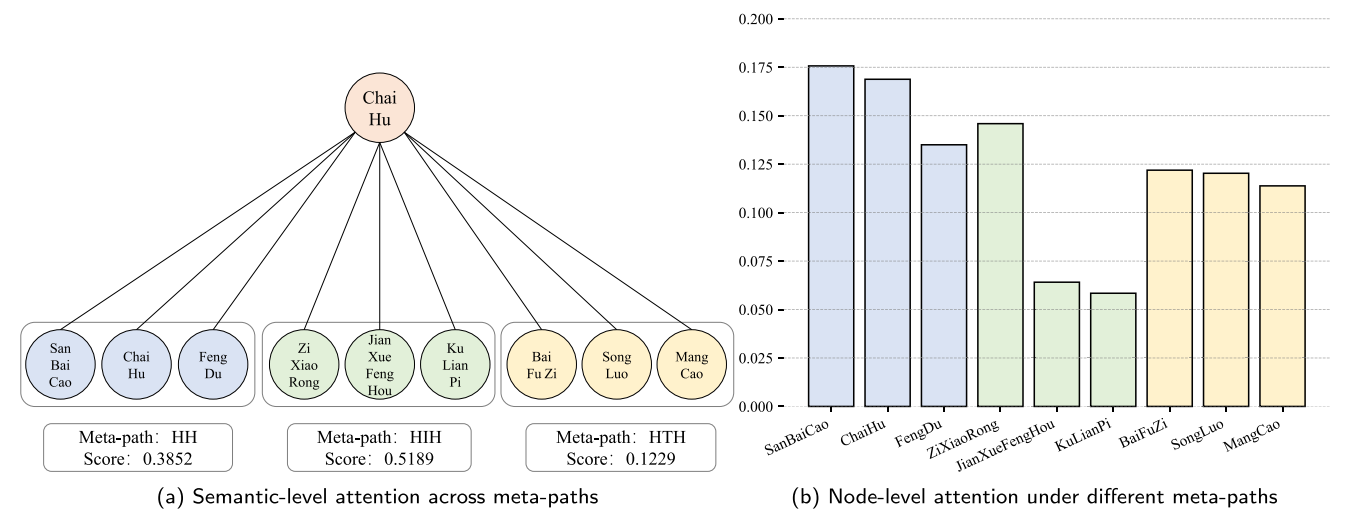


图 2. Hierarchical attention analysis results of HerbToxNet on the herb node *Chai Hu* including (a) semantic-level attention across meta-paths and (b) node-level attention under different meta-paths.

Fig. 2. Hierarchical attention analysis results of HerbToxNet on the herb node *Chai Hu*, including (a) semantic-level attention across meta-paths, and (b) node-level attention under different meta-paths.

进一步 表明 节点 自身 在 学习 过程中 对 生成 其 自身 表征 起着 关键 作用 。

在HIH元路径中，柴胡与紫霄榕、见血封喉和苦楝皮相连。柴胡、紫霄榕、苦楝皮和见血封喉共享共同的次级代谢产物类别，包括三萜皂苷、黄酮类化合物、酚类化合物、挥发油和苦味生物碱。例如，柴胡含有柴胡皂苷，紫霄榕富含黄酮类化合物和酚酸，苦楝皮含有川楝素及相关柠檬苦素类化合物，而见血封喉则含有强心苷类毒性成分如见血封喉苷。尽管这些成分在结构上具有多样性，但它们共享重叠的生物活性，特别是抗炎、抗肿瘤和解毒作用。

在HTH元路径中，柴胡与白附子、松萝和芒草相连。这些草药与柴胡共享若干潜在药理学靶点，包括肿瘤坏死因子、白细胞介素6、前列腺素内过氧化物合酶2和丝裂原活化蛋白激酶1，这些靶点与肝损伤和炎症反应密切相关。柴胡中的柴胡皂苷和白附子中的二萜类生物碱可能通过激活NFKB信号通路上调促炎细胞因子（如肿瘤坏死因子和白细胞介素6）。松萝中的松萝酸与线粒体功能障碍和氧化应激有关，导致前列腺素内过氧化物合酶2过表达介导的肝毒性。芒草中的生物碱可能影响丝裂原活化蛋白激酶1信号传导，该通路参与肝细胞凋亡和应激反应。

总之，节点级 注意力 机制 基于共享的 草药 属性、成分 和 靶点 突出显示了关键关联，为模型对柴胡的 预测 提供了具有生物学意义且可解释的见解。

4.7.2. 生物可解释性 分析

为增强HerbToxNet的生物可解释性，我们沿“草药-成分-靶点-生物通路-毒性”轴线构建了多层次解释框架。以柴胡为代表案例，通过整合元路径分析、文献整理和毒性数据库，我们系统识别了其毒性成分、相关分子靶点及关联毒性类型。我们进一步通过生物通路和功能注释将这些组分相互关联。结果表明，HerbToxNet成功揭示了参与肝毒性、肾毒性和全身毒性的关键毒性成分及其分子靶点，并揭示了中药诱导毒性反应的相关生物通路和机制。这证明HerbToxNet能为其毒性预测提供机制上有意义且生物学上合理的解释，从而支持更可靠的草药风险评估。分析流程详解、深度结果及所有支撑证据详见补充章节V。

5. 结论

本文构建了一个多标签草药毒性数据集，并提出HerbToxNet——一个融合异质网络建模、对比学习和加权标签融合的草药毒性预测框架。实证研究表明，HerbToxNet相比竞争方法实现了更优的性能。该框架能够识别新型潜在毒性，提升草药的临床用药安全性，同时可揭示毒性相关成分与靶点，为毒性机制研究提供见解，指导解毒剂开发，支持毒性相关数据库建设及更安全草药的发现。尽管HerbToxNet在性能上优于对比方法并提供了可靠结果，但仍存在局限。当前数据集覆盖五种毒性类别，多数草药被赋予多重标签，这支持了多标签分类的应用。由于标准化数据有限，现阶段未纳入更广泛的毒性类别。未来工作中，我们将扩展毒性覆盖范围以增强HerbToxNet的鲁棒性与适用性。

CRediT 作者 贡献 声明

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数据 可用性

我们的 代码 与 收集的 数据集 已 共享于 <https://www.sdu-idea.cn/codes.php?name=HerbToxNet>。

利益冲突 声明

于国贤报告由中华人民共和国科学技术部提供资助。若存在其他作者，他们声明无任何可能影响本文报告成果的已知竞争性财务利益或个人关系。

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补充 材料

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参考文献

Caesar, L., & Cech, N. (2019). Synergy and antagonism in natural product extracts: when 1 + 1 does not equal 2. *Natural Product Reports*, 36(6), 869–888. <https://doi.org/10.1039/C9NP00011A>.
K. (2003). Some aspects of toxic contaminants in herbal medicines. *Chemosphere*, 48(9), 1361–1371. [https://doi.org/10.1016/S0045-6535\(03\)00471-5](https://doi.org/10.1016/S0045-6535(03)00471-5).
& Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *Int. conf. mach. learn.* (pp. 1597–1607). Cherkasov, A., Muratov, E., Fourches, D., Varnek, A., Baskin, I., Cronin, M., Dearden, J., Gramatica, P., Martin, Y., Todeschini, R., et al. (2014). Qsar modeling: Where have you been? where are you going to? *Journal of Medicinal Chemistry*, 57(12), 4977–5010. <https://doi.org/10.1021/jm4004285>.
S. Dong, L. Liu, L. Guo, J. Zhao, L. Zhang, J. Bu, D. Liu, X. Huo, P. Cao, W. et al. (2021). Herb: A high-throughput experiment-and reference-guided database of traditional chinese medicine. *Nucleic Acids Research*, 49(D1), D1197–D1206. <https://doi.org/10.1093/nar/gkaa1063>.
Flora, S., Shrivastava, R., & Mittal, M. (2013). Chemistry and pharmacological properties of some natural and synthetic antioxidants for heavy metal toxicity. *Current Medicinal Chemistry*, 20(36), 4540–4574. <https://doi.org/10.2174/09298673113209990146>.
Gao, J., Wu, L., Lin, G., Zou, J., Yan, B., Liu, K., He, S., & Bo, X. (2025). Multi-task multi-view and iterative error-correcting random forest for acute toxicity prediction. *Expert Systems with Applications*, 274, 126972. <https://doi.org/10.1016/j.eswa.2025.126972>.
Y. Liang, A. Fan, X. Hu, L. Hao, F. & Li, Y. (2019). Safety research in traditional chinese medicine: methods, applications and outlook. *Engineering*, 5(1), 76–82. <https://doi.org/10.1016/j.eng.2018.11.019>.
Ge, Z., Jiang, X., Tong, Z., Feng, P., Zhou, B., Xu, M., Wang, Z., & Pang, Y. (2021). Multi-label correlation guided feature fusion network for abnormal ECG diagnosis. *Knowledge-Based System*, 233, 107508. <https://doi.org/10.1016/j.knosys.2021.107508>.
L. Jiang, J. Chen, S. & Qi, M. (2023). A syndrome differentiation model of TCM based on multi-label deep forest using biomedical text mining. *Frontiers in Genetics*, 14, 1272016. <https://doi.org/10.3389/fgene.2023.1272016>.
Y. (2006). Dimensionality reduction by learning an invariant mapping. In *IEEE conf. comput. vis. pattern recognit. (cvpr)* (pp. 1735–1742). <https://doi.org/10.1109/CVPR.2006.100>.
Hadsell, R., Chopra, S., & LeCun, Y. (2006). Dimensionality reduction by learning an invariant mapping. In *IEEE conf. comput. vis. pattern recognit. (cvpr)* (pp. 1735–1742). <https://doi.org/10.1109/CVPR.2006.100>.

further indicates that the node itself plays a crucial role in generating its own representation during the learning process.

For HIH meta-path, *Chai Hu* is connected to *Zi Xiao Rong*, *Jian Xue Feng Hou*, and *Ku Lian Pi*. *Chai Hu*, *Zi Xiao Rong*, *Ku Lian Pi*, and *Jian Xue Feng Hou* share common classes of secondary metabolites, which include triterpenoid saponins, flavonoids, phenolic compounds, volatile oils, and bitter alkaloids. For instance, *Chai Hu* contains saikosaponins, *Zi Xiao Rong* is rich in flavonoids and phenolic acids, *Ku Lian Pi* contains toosendanin and related limonoids, while *Jian Xue Feng Hou* harbors cardiotoxic cardiac glycosides such as antiarin. Despite their structural diversity, these ingredients share overlapped biological activities, notably anti-inflammatory, antitumor, and detoxifying effects.

For the HTH meta-path, *Chai Hu* is connected to *Bai Fu Zi*, *Song Luo*, and *Mang Cao*. These herbs share several potential pharmacological targets with *Chai Hu*, including *TNF*, *IL6*, *PTGS2*, and *MAPK1*, which are closely associated with liver injury and inflammatory responses. Saikosaponins in *Chai Hu* and diterpenoid alkaloids in *Bai Fu Zi* may upregulate pro-inflammatory cytokines such as *TNF* and *IL6* via activation of the *NFKB* signaling pathway. Usnic acid in *Song Luo* has been implicated in mitochondrial dysfunction and oxidative stress, leading to hepatotoxicity mediated by *PTGS2* overexpression. Alkaloids in *Mang Cao* may affect *MAPK1* signaling, which is involved in hepatocyte apoptosis and stress response.

In summary, the node-level attention mechanism highlights key associations based on shared herb properties, ingredients, and targets, offering biologically meaningful and interpretable insights into the model’s predictions for *Chai Hu*.

4.7.2. Biointerpretability analysis

To enhance the biological interpretability of HerbToxNet, we constructed a multi-level explanatory framework along the “herb-ingredient-target-biological pathway-toxicity” axis. Taking *Chai Hu* as a representative case, we systematically identified its toxic ingredients, related molecular targets, and associated toxicity types by integrating meta-path analysis, literature curation, and toxicity databases. We further linked these components via their biological pathways and functional annotations. Our results show that HerbToxNet successfully uncovers key toxic ingredients and their molecular targets involved in hepatotoxicity, nephrotoxicity, and systemic toxicity, and reveals the relevant biological pathways and mechanisms underlying herb-induced toxic responses. This demonstrates that HerbToxNet can provide mechanistically meaningful and biologically plausible explanations for its toxicity predictions, thereby supporting more reliable risk assessment of herbs. A comprehensive description of the analytical procedures, in-depth results, and all supporting evidence can be found in Supplementary Section V.

5. Conclusion

In this paper, we construct a multi-label herb toxicity dataset and propose HerbToxNet, a framework for herb toxicity prediction that integrates heterogeneous network modeling, contrastive learning, and weighted label fusion. Empirical study shows that HerbToxNet achieves superior performance than competitive methods. HerbToxNet can identify novel potential toxicities and improve the safe clinical use of herbs. Moreover, it can uncover toxicity-related ingredients and targets, provide insights into toxicity mechanisms, guide antidote development, support toxicity-related databases construction and the discovery of safer herbs. Although HerbToxNet achieves better performance than compared methods and gives some authentic results, it still has limitations. Our dataset currently covers five toxicity categories, with many herbs assigned multiple labels, supporting the use of multi-label classification. Due to limited standardized data, broader toxicity categories are not included at this stage. In future work, we will expand toxicity coverage to enhance the robustness and applicability of HerbToxNet.

CRediT authorship contribution statement

Yongzheng Zhu: Data curation, Formal analysis, Investigation, Software, Validation, Writing – original draft; **Yunbo Miao**: Formal analysis, Software, Validation, Writing – original draft; **Rong Sun**: Supervision, Resources; **Zhongmin Yan**: Methodology, Formal analysis, Writing – review & editing; **Guoxian Yu**: Supervision, Investigation, Methodology, Formal analysis, Writing – review & editing.

Data Availability

Our code and collected dataset are shared at <https://www.sdu-idea.cn/codes.php?name=HerbToxNet>.

Declaration of competing interest

Guoxian Yu reports financial support was provided by Ministry of Science and Technology of the People’s Republic of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.eswa.2025.129969](https://doi.org/10.1016/j.eswa.2025.129969).

References

Caesar, L., & Cech, N. (2019). Synergy and antagonism in natural product extracts: when 1 + 1 does not equal 2. *Natural Product Reports*, 36(6), 869–888. <https://doi.org/10.1039/C9NP00011A>.
Chan, K. (2003). Some aspects of toxic contaminants in herbal medicines. *Chemosphere*, 52(9), 1361–1371. [https://doi.org/10.1016/S0045-6535\(03\)00471-5](https://doi.org/10.1016/S0045-6535(03)00471-5).
Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *Int. conf. mach. learn.* (pp. 1597–1607). <https://doi.org/10.5555/3524938.3525087>.
Cherkasov, A., Muratov, E., Fourches, D., Varnek, A., Baskin, I., Cronin, M., Dearden, J., Gramatica, P., Martin, Y., Todeschini, R. et al. (2014). Qsar modeling: Where have you been? where are you going to? *Journal of Medicinal Chemistry*, 57(12), 4977–5010. <https://doi.org/10.1021/jm4004285>.
Fang, S., Dong, L., Liu, L., Guo, J., Zhao, L., Zhang, J., Bu, D., Liu, X., Huo, P., Cao, W. et al. (2021). Herb: A high-throughput experiment-and reference-guided database of traditional chinese medicine. *Nucleic Acids Research*, 49(D1), D1197–D1206. <https://doi.org/10.1093/nar/gkaa1063>.
Flora, S., Shrivastava, R., & Mittal, M. (2013). Chemistry and pharmacological properties of some natural and synthetic antioxidants for heavy metal toxicity. *Current Medicinal Chemistry*, 20(36), 4540–4574. <https://doi.org/10.2174/09298673113209990146>.
Gao, J., Wu, L., Lin, G., Zou, J., Yan, B., Liu, K., He, S., & Bo, X. (2025). Multi-task multi-view and iterative error-correcting random forest for acute toxicity prediction. *Expert Systems with Applications*, 274, 126972. <https://doi.org/10.1016/j.eswa.2025.126972>.
Gao, Y., Liang, A., Fan, X., Hu, L., Hao, F., & Li, Y. (2019). Safety research in traditional chinese medicine: methods, applications, and outlook. *Engineering*, 5(1), 76–82. <https://doi.org/10.1016/j.eng.2018.11.019>.
Ge, Z., Jiang, X., Tong, Z., Feng, P., Zhou, B., Xu, M., Wang, Z., & Pang, Y. (2021). Multi-label correlation guided feature fusion network for abnormal ECG diagnosis. *Knowledge-Based System*, 233, 107508. <https://doi.org/10.1016/j.knosys.2021.107508>.
Gong, L., Jiang, J., Chen, S., & Qi, M. (2023). A syndrome differentiation model of TCM based on multi-label deep forest using biomedical text mining. *Frontiers in Genetics*, 14, 1272016. <https://doi.org/10.3389/fgene.2023.1272016>.
Hadsell, R., Chopra, S., & LeCun, Y. (2006). Dimensionality reduction by learning an invariant mapping. In *IEEE conf. comput. vis. pattern recognit. (cvpr)* (pp. 1735–1742). <https://doi.org/10.1109/CVPR.2006.100>.

He, S., Zhang, C., Zhou, P., Zhang, X., Ye, T., Wang, R., Sun, G., & Sun, X. (2019a). Herb-induced liver injury: Phylogenetic relationship structure-toxicity relationship and herb-ingredient network analysis. *International Journal of Molecular Sciences*, 20(15), 3633. <https://doi.org/10.3390/ijms20153633>.

S. Zhang X. Lu S. Zhu T. Sun G. & Sun X. (2019b). A computational toxicology approach to screen the hepatotoxic ingredients in traditional chinese medicines: Polymultiflorum thunb as a case study. *Biomolecules* 9(10) 577. <https://doi.org/10.3390/10.3390/10.3390/ijms20153633>.

H. Li Y. Zheng Z. Hu W. Lin R. & Kang Y. (2025). A traditional chinese medicine prescription recommendation model based on contrastive pre-training and hierarchical structure network. *Expert Systems with Applications* 268 126318. <https://doi.org/10.1016/j.eswa.2024.126318>.

C. Li X. Hu S. Liu G. Fang J. Zhou X. Yan X. & Yan B. (2024). Advanced mass-spectra-based machine learning for predicting the toxicity of traditional chinese medicines. *Analytical Chemistry*, 97(1), 783–792. <https://doi.org/10.1021/acs.analchem.4c05311>.

M. Yang J. Zhang C. Liu B. Chan K. Cao H. & Lu A. (2010). Clinical studies with traditional chinese medicine in the past decade and future research and development. *Planta Medica*, 76(17), 2048–2064. <https://doi.org/10.1055/s-0030-1250456>.

A. Trifan A. Vieriu M. Caba I.-C. Mărtu I. & Agoroei L. (2022). Current trends in toxicity assessment of herbal medicines: A narrative review. *Processes* 11(1) 83. <https://doi.org/10.3390/pr11010083>.

P. Teterwak P. Wang C. Sarna A. Tian Y. Isola P. Maschinot A. Liu C. & Krishnan D. (2020). Supervised contrastive learning. In *Adv. neural inf. process. syst.* (pp. 18661–18673). <https://doi.org/10.5555/3495724.3497291>.

Liu, Z., & Thakkar, S. (2020). DeepDILI: Deep learning-driven drug-induced liver injury prediction using model-level representation. *Chemical Research in Toxicology* 34(2) 550–565. <https://doi.org/10.1021/acs.chemrestox.0c00374>.

Y. Zhang Y. Wang Y. Li Y. Yang F. Zhang P. Zhang Y. & Liu C. (2019). A strategy for the discovery and validation of toxicity quality marker of chinese medicine based on network toxicology. *Phytomedicine* 54 365–370. <https://doi.org/10.1016/j.phymed.2018.01.018>.

Liu, R., Li X., Huang, N., Fan, M., & Sun, R. (2020). Toxicity of traditional Chinese medicine herbal and mineral products. *Advances in Pharmacology* 87 301–346. <https://doi.org/10.1016/j.bs.apha.2019.08.001>.

Q. Chen G. He H. Yang Z. Zhao L. Zhang K. & Chen C.Y.C. (2023). Tcmbank-the largest tcm database provides deep learning-based chinese-western medicine exclusion prediction. *Signal Transduction and Targeted Therapy* 8(1) 127. <https://doi.org/10.1016/j.slct.2023.100374>.

Yong, J., Li, Y., & Xie, H. (2022). Graph-based multi-label disease prediction model learning from medical data and domain knowledge. *Knowledge-Based Systems*, 235, 107662. <https://doi.org/10.1016/j.knsys.2021.107662>.

Y. Zhang Y. Peng S. Huo M. Li J. Cao Y. & Chen Z. (2022). Property theory of chinese materia medica: Clinical pharmacodynamics of traditional chinese medicine. *Journal of Traditional Chinese Medical Sciences* 9(1) 7–12. <https://doi.org/10.1016/j.jtcms.2022.01.006>.

Qiu, S., Wang M., Yang Y., Yu G., Wang J., Yan Z., Domeniconi C., & Guo M. (2023). Meta multi-instance multi-label learning by heterogeneous network fusion. *An International Journal on Information Fusion* 94 272–283. <https://doi.org/10.1016/j.inffus.2023.02.010>.

& Bajic, V. (2016). In silico toxicology: Computational methods for the prediction of chemical toxicity. *Wiley Interdisciplinary Reviews Computational Molecular Science* 6(2) 147–172. <https://doi.org/10.1002/wcms.1240>.

T. Ben-Baruch E. Zamir N. Noy A. Friedman I. Protter M. & Zelnik-Manor L. (2021). Asymmetric loss for multi-label classification. In *IEEE/CVF int. conf. comput. vis.* (pp. 82–91). <https://doi.org/10.1109/ICCV48922.2021.00015>.

J. Swanson K. Simon E. Zhang A. Ghazizadeh Z. Ogden S. Hamadeh H. & Zou J. (2024). Unitox: Leveraging LLMs to curate a unified dataset of drug-toxicity from FDA labels. *Advances in Neural Information Processing Systems* 37 12078–12093. <https://doi.org/10.1101/2024.06.21.24309315>.

孙, Y. Shi S. Li Y. & Wang Q. (2019). Development of quantitative structure-activity relationship models to predict potential nephrotoxic ingredients in traditional chinese medicines. *Food and Chemical Toxicology* 128 163–170. <https://doi.org/10.1016/j.fct.2019.03.056>.

王, X. Ji H. Shi C. Wang B. Ye Y. Cui P. & Yu P. (2019). Heterogeneous graph attention network. In *Www conf.* (pp. 2022–2032). <https://doi.org/10.1145/3308558.3313562>.

X. Qu H. Liu P. & Cheng Y. (2004). A self-learning expert system for diagnosis in traditional chinese medicine. *Expert Systems with Applications* 26(4) 557–566. <https://doi.org/10.1016/j.eswa.2003.10.004>.

Z. Li L. Yan J. & Yao Y. (2020). Approaching high-accuracy side effect prediction of traditional chinese medicine compound prescription using network embedding and deep learning. *IEEE Access* 8 82493–82499. <https://doi.org/10.1109/ACCESS.2020.2991750>.

J. Dai J. Sun Y. Meng Z. Ma H. & Zhou Y. (2024). Tirpnet: Risk prediction of traditional chinese medicine ingredients based on a deep neural network. *Journal of Ethnopharmacology* 325 117860. <https://doi.org/10.1016/j.jep.2024.117860>.

Q. Cai C. Guo P. Chen M. Wu X. Zhou J. Luo Y. Zou Y. Liu A. Wang Q. et al. (2019a). In silico identification and mechanism exploration of hepatotoxic ingredients in traditional chinese medicine. *Frontiers in Pharmacology* 10 458. <https://doi.org/10.3389/fphar.2019.00458>.

Q. Yin C. Li Y. Cai J. Yang H. Huang Y. Zheng Y. Xiong K. Yu H. Lu A. et al. (2021). Detecting critical functional ingredients group and mechanism of xuebijing injection in treating sepsis. *Frontiers in Pharmacology* 12 769190. <https://doi.org/10.3389/fphar.2021.769190>.

Y. Zhang F. Yang K. Fang S. Bu D. Li H. Sun L. Hu H. Gao K. Wang W. et al. (2019b). SymMap: An integrative database of traditional chinese medicine enhanced by symptom mapping. *Nucleic Acids Research* 47(D1) 10193–10211. <https://doi.org/10.1093/nar/gky1021>.

X. Xu H. Shang Y. Zhu R. Hong X. Song Z. & Yang Z. (2021). Development of the general chapters of the chinese pharmacopoeia 2020 edition: A review. *Journal of Pharmaceutical Analysis* 11(4) 398–404. <https://doi.org/10.1016/j.jpba.2021.05.001>.

J., Wu, L., He, S., Liu, K., & Bo, X. (2022a). An algorithm framework for drug-induced liver injury prediction based on genetic algorithm and ensemble learning. *Molecules*, 27(10), 3112. <https://doi.org/10.3390/10.3390/10.3390/ijms20153633>.

D. Zheng G. Wang C. Chen Z. Mao T. Gao J. Yan Y. Chen X. Ji X. Yu J. et al. (2022b). Hit 2.0: An enhanced platform for herbal ingredients' targets. *Nucleic Acids Research* 50(D1) D1238–D1243. <https://doi.org/10.1093/nar/gkab1011>.

J. Wang S. Zhang T. Sun Y. Han L. Banahene P. O. & Wang Q. (2021). Predicting the potential toxicity of 26 components in cassiae semen using in silico and in vitro approaches. *Current Research in Toxicology* 2 237–245. <https://doi.org/10.1016/j.crtox.2021.100374>.

1016 2021 06 001

K. Zhou X. Zhang R. Liu B. Lei L. Zhang X. Chu H. Sun C. Gao Z. & Xu H. (2015). Integrating herb effect similarity for network-based herb target prediction. In *Int. conf. biomed. eng. inform.* (pp. 483–488). <https://doi.org/10.1109/BMEI.2015.7401553>.

Z. Kou F. Gao Y. Gao F. Lyu C.-m. & Wei H. (2025). A machine learning model for predicting abnormal liver function induced by a Chinese herbal medicine preparation (Zhengqing fengtongning) in patients with rheumatoid arthritis based on real-world study. *Journal of Integrative Medicine* 23(1) 25–35. <https://doi.org/10.1016/j.joim.2025.02.001>.

2024 12 001

Y. Ren L. Sun R. Wang J. & Yu G. (2025). Herb-target interaction prediction by multi-instance learning. *IEEE Transactions on Artificial Intelligence* 6(5) 1184–1193. <https://doi.org/10.1109/TAI.2024.3515932>.

He, S., Zhang, C., Zhou, P., Zhang, X., Ye, T., Wang, R., Sun, G., & Sun, X. (2019a). Herb-induced liver injury: Phylogenetic relationship, structure-toxicity relationship, and herb-ingredient network analysis. *International Journal of Molecular Sciences*, 20(15), 3633. <https://doi.org/10.3390/ijms20153633>.

He, S., Zhang, X., Lu, S., Zhu, T., Sun, G., & Sun, X. (2019b). A computational toxicology approach to screen the hepatotoxic ingredients in traditional chinese medicines: Polygonum multiflorum thunb as a case study. *Biomolecules*, 9(10), 577. <https://doi.org/10.3390/biom9100577>.

Hu, H., Li, Y., Zheng, Z., Hu, W., Lin, R., & Kang, Y. (2025). A traditional chinese medicine prescription recommendation model based on contrastive pre-training and hierarchical structure network. *Expert Systems with Applications*, 268, 126318. <https://doi.org/10.1016/j.eswa.2024.126318>.

Jia, C., Li, X., Hu, S., Liu, G., Fang, J., Zhou, X., Yan, X., & Yan, B. (2024). Advanced mass-spectra-based machine learning for predicting the toxicity of traditional chinese medicines. *Analytical Chemistry*, 97(1), 783–792. <https://doi.org/10.1021/acs.analchem.4c05311>.

Jiang, M., Yang, J., Zhang, C., Liu, B., Chan, K., Cao, H., & Lu, A. (2010). Clinical studies with traditional chinese medicine in the past decade and future research and development. *Planta Medica*, 76(17), 2048–2064. <https://doi.org/10.1055/s-0030-1250456>.

Jităreanu, A., Trifan, A., Vieriu, M., Caba, I.-C., Mărtu, I., & Agoroei, L. (2022). Current trends in toxicity assessment of herbal medicines: A narrative review. *Processes*, 11(1), 83. <https://doi.org/10.3390/pr11010083>.

Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., & Krishnan, D. (2020). Supervised contrastive learning. In *Adv. neural inf. process. syst.* (pp. 18661–18673). <https://doi.org/10.5555/3495724.3497291>.

Li, T., Tong, W., Roberts, R., Liu, Z., & Thakkar, S. (2020). DeepDILI: Deep learning-powered drug-induced liver injury prediction using model-level representation. *Chemical Research in Toxicology*, 34(2), 550–565. <https://doi.org/10.1021/acs.chemrestox.0c00374>.

Li, Y., Zhang, Y., Wang, Y., Li, Y., Yang, F., Zhang, P., Zhang, Y., & Liu, C. (2019). A strategy for the discovery and validation of toxicity quality marker of chinese medicine based on network toxicology. *Phytomedicine*, 54, 365–370. <https://doi.org/10.1016/j.phymed.2018.01.018>.

Liu, R., Li, X., Huang, N., Fan, M., & Sun, R. (2020). Toxicity of traditional Chinese medicine herbal and mineral products. *Advances in Pharmacology*, 87, 301–346. <https://doi.org/10.1016/j.bs.apha.2019.08.001>.

Lv, Q., Chen, G., He, H., Yang, Z., Zhao, L., Zhang, K., & Chen, C. Y. C. (2023). Tcmbank-the largest tcm database provides deep learning-based chinese-western medicine exclusion prediction. *Signal Transduction and Targeted Therapy*, 8(1), 127. <https://doi.org/10.1038/s41392-023-01339-1>.

Pham, T., Tao, X., Zhang, J., Yong, J., Li, Y., & Xie, H. (2022). Graph-based multi-label disease prediction model learning from medical data and domain knowledge. *Knowledge-Based System*, 235, 107662. <https://doi.org/10.1016/j.knsys.2021.107662>.

Qiao, Y., Zhang, Y., Peng, S., Huo, M., Li, J., Cao, Y., & Chen, Z. (2022). Property theory of chinese materia medica: Clinical pharmacodynamics of traditional chinese medicine. *Journal of Traditional Chinese Medical Sciences*, 9(1), 7–12. <https://doi.org/10.1016/j.jtcms.2022.01.006>.

Qiu, S., Wang, M., Yang, Y., Yu, G., Wang, J., Yan, Z., Domeniconi, C., & Guo, M. (2023). Meta multi-instance multi-label learning by heterogeneous network fusion. *An International Journal on Information Fusion*, 94, 272–283. <https://doi.org/10.1016/j.inffus.2023.02.010>.

Raies, A., & Bajic, V. (2016). In silico toxicology: Computational methods for the prediction of chemical toxicity. *Wiley Interdisciplinary Reviews Computational Molecular Science*, 6(2), 147–172. <https://doi.org/10.1002/wcms.1240>.

Ridnik, T., Ben-Baruch, E., Zamir, N., Noy, A., Friedman, I., Protter, M., & Zelnik-Manor, L. (2021). Asymmetric loss for multi-label classification. In *IEEE/CVF int. conf. comput. vis.* (pp. 82–91). <https://doi.org/10.1109/ICCV48922.2021.00015>.

Silberg, J., Swanson, K., Simon, E., Zhang, A., Ghazizadeh, Z., Ogden, S., Hamadeh, H., & Zou, J. (2024). Unitox: Leveraging LLMs to curate a unified dataset of drug-induced toxicity from FDA labels. *Advances in Neural Information Processing Systems*, 37, 12078–12093. <https://doi.org/10.1101/2024.06.21.24309315>.

Sun, Y., Shi, S., Li, Y., & Wang, Q. (2019). Development of quantitative structure-activity relationship models to predict potential nephrotoxic ingredients in traditional chinese medicines. *Food and Chemical Toxicology*, 128, 163–170. <https://doi.org/10.1016/j.fct.2019.03.056>.

Wang, X., Ji, H., Shi, C., Wang, B., Ye, Y., Cui, P., & Yu, P. (2019). Heterogeneous graph attention network. In *Www conf.* (pp. 2022–2032). <https://doi.org/10.1145/3308558.3313562>.

Wang, X., Qu, H., Liu, P., & Cheng, Y. (2004). A self-learning expert system for diagnosis in traditional chinese medicine. *Expert Systems with Applications*, 26(4), 557–566. <https://doi.org/10.1016/j.eswa.2003.10.004>.

Wang, Z., Li, L., Yan, J., & Yao, Y. (2020). Approaching high-accuracy side effect prediction of traditional chinese medicine compound prescription using network embedding and deep learning. *IEEE Access*, 8, 82493–82499. <https://doi.org/10.1109/ACCESS.2020.2991750>.

Wei, J., Dai, J., Sun, Y., Meng, Z., Ma, H., & Zhou, Y. (2024). Tirpnet: Risk prediction of traditional chinese medicine ingredients based on a deep neural network. *Journal of Ethnopharmacology*, 325, 117860. <https://doi.org/10.1016/j.jep.2024.117860>.

Wu, Q., Cai, C., Guo, P., Chen, M., Wu, X., Zhou, J., Luo, Y., Zou, Y., Liu, A., Wang, Q. et al. (2019a). In silico identification and mechanism exploration of hepatotoxic ingredients in traditional chinese medicine. *Frontiers in Pharmacology*, 10, 458. <https://doi.org/10.3389/fphar.2019.00458>.

Wu, Q., Yin, C., Li, Y., Cai, J., Yang, H., Huang, Y., Zheng, Y., Xiong, K., Yu, H., Lu, A. et al. (2021). Detecting critical functional ingredients group and mechanism of xuebijing injection in treating sepsis. *Frontiers in Pharmacology*, 12, 769190. <https://doi.org/10.3389/fphar.2021.769190>.

Wu, Y., Zhang, F., Yang, K., Fang, S., Bu, D., Li, H., Sun, L., Hu, H., Gao, K., Wang, W. et al. (2019b). SymMap: An integrative database of traditional chinese medicine enhanced by symptom mapping. *Nucleic Acids Research*, 47(D1), D1110–D1117. <https://doi.org/10.1093/nar/gky1021>.

Xu, X., Xu, H., Shang, Y., Zhu, R., Hong, X., Song, Z., & Yang, Z. (2021). Development of the general chapters of the chinese pharmacopoeia 2020 edition: A review. *Journal of Pharmaceutical Analysis*, 11(4), 398–404. <https://doi.org/10.1016/j.jpba.2021.05.001>.

Yan, B., Ye, X., Wang, J., Han, J., Wu, L., He, S., Liu, K., & Bo, X. (2022a). An algorithm framework for drug-induced liver injury prediction based on genetic algorithm and ensemble learning. *Molecules*, 27(10), 3112. <https://doi.org/10.3390/10.3390/10.3390/ijms20153633>.

Yan, D., Zheng, G., Wang, C., Chen, Z., Mao, T., Gao, J., Yan, Y., Chen, X., Ji, X., Yu, J. et al. (2022b). Hit 2.0: An enhanced platform for herbal ingredients' targets. *Nucleic Acids Research*, 50(D1), D1238–D1243. <https://doi.org/10.1093/nar/gkab1011>.

Yang, J., Wang, S., Zhang, T., Sun, Y., Han, L., Banahene, P. O., & Wang, Q. (2021). Predicting the potential toxicity of 26 components in cassiae semen using in silico and in vitro approaches. *Current Research in Toxicology*, 2, 237–245. <https://doi.org/10.1016/j.crtox.2021.06.001>.

Yang, K., Zhou, X., Zhang, R., Liu, B., Lei, L., Zhang, X., Chu, H., Sun, C., Gao, Z., & Xu, H. (2015). Integrating herb effect similarity for network-based herb target prediction. In *Int. conf. biomed. eng. inform.* (pp. 483–488). <https://doi.org/10.1109/BMEI.2015.7401553>.

Yu, Z., Kou, F., Gao, Y., Gao, F., Lyu, C.-m., & Wei, H. (2025). A machine learning model for predicting abnormal liver function induced by a Chinese herbal medicine preparation (Zhengqing fengtongning) in patients with rheumatoid arthritis based on real-world study. *Journal of Integrative Medicine*, 23(1), 25–35. <https://doi.org/10.1016/j.joim.2024.12.001>.

Zhu, Y., Ren, L., Sun, R., Wang, J., & Yu, G. (2025). Herb-target interaction prediction by multi-instance learning. *IEEE Transactions on Artificial Intelligence*, 6(5), 1184–1193. <https://doi.org/10.1109/TAI.2024.3515932>.